FIRST CENTRAL MOMENT IN THE EVEN HYPERELLIPTIC ENSEMBLE WITH AN EXPLICIT MINIMAL-PERIOD-THREE SECONDARY TERM

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Allowed standard prerequisites (editorial).

- Quadratic function-field L-functions, completed central values, Poisson/Gauss sums and contour analysis. Original source-local statements/derivations and full amsrefs bibliography retained.

1. Introduction

Let q be an odd prime power. We study the first moment of quadratic Dirichlet L-functions at the central point over the even hyperelliptic ensemble,

$$M(g) := \sum_{D\in\mathcal{H}_{2g+2}} L\big(\tfrac{1}{2},\chi_D\big)\,, \tag{1.1}$$

as $g\to\infty$ with q fixed. Here $\mathcal{H}_{2g+2}$ is the set of monic square-free polynomials of degree $2g+2$ in $\mathbb{F}_q[x]$, and χ_D is the primitive quadratic character attached to D (see §2.1).

For the odd hyperelliptic ensemble $\mathcal{H}_{2g+1}$, the first moment was obtained by Andrade and Keating [1], who also formulated conjectures for the higher moments [2] following the recipe of Conrey, Farmer, Keating, Rubinstein and Snaith [5]. Florea [6] identified a secondary term of order $gq^{(2g+1)/3}$ and reduced the error to $O_\varepsilon(q^{g(1+\varepsilon)/2})$; see [7] for the second and third moments and [8] for the fourth. For the even hyperelliptic ensemble $\mathcal{H}_{2g+2}$, Jung [10] and Andrade–MacMillan [3] obtained the principal term and the same error exponent, with lower-order terms of order $gq^{2g/3}$ whose coefficients do not depend on g mod 3. The results of [6], [10] and [3] are all stated for prime q with $q\equiv 1 \bmod 4$. For an arbitrary odd prime power q, it is shown in [11] that the complete secondary contribution over $\mathcal{H}_{2g+1}$ is $q^{2g/3}\big(\alpha_g g+\beta_g\big)$, whose dependence on g has minimal period exactly three.

The central idea of this paper is to work with the *completed L-function*

$$L^*(s,\chi_D) := \frac{L(s,\chi_D)}{1-q^{-s}}$$

in place of $L(s,\chi_D)$ itself: for $D\in\mathcal{H}_{2g+2}$ the character χ_D is even, so $L(s,\chi_D)$ carries the trivial factor $1-q^{-s}$, and $L^*(s,\chi_D)$ has a symmetric functional equation (see §2.1). We therefore introduce the associated completed first moment

$$M^*(g) := \sum_{D\in\mathcal{H}_{2g+2}} L^*\big(\tfrac{1}{2},\chi_D\big)\,. \tag{1.2}$$

At the central point one has $L\big(\frac{1}{2},\chi_D\big) = (1-q^{-1/2})L^*\big(\frac{1}{2},\chi_D\big)$, and summing this identity over $D\in\mathcal{H}_{2g+2}$ gives

$$M(g) = (1-q^{-1/2})M^*(g). \tag{1.3}$$

The analysis is accordingly carried out for $M^*(g)$, and $M(g)$ is recovered from (1.3) at the end.

A second departure from the earlier treatments concerns the ground field. Neither a primality nor a congruence condition is imposed on q: throughout this paper q is an arbitrary odd prime power. The obstruction is that in general the quadratic Gauss sums over $\mathbb{F}_q[x]$ carry a root-of-unity phase, which propagates into the Poisson summation formula built from them. We therefore work with a phase-normalized Gauss sum, into whose definition that phase is absorbed, and with the Poisson summation formula it yields; both are free of the phase for every odd prime power q (see §2.3; cf. [6, Section 9]).

Let $\zeta_{\mathcal{A}}$ denote the zeta function of $\mathcal{A}:=\mathbb{F}_q[x]$, so that $\zeta_{\mathcal{A}}(s)=\big(1-q^{1-s}\big)^{-1}$ (see §2.1), and

$$\mathscr{P}(s) := \prod_P \left(1-\frac{1}{|P|^s(|P|+1)}\right),$$

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where the product is taken over all monic irreducible polynomials $P \in \mathcal{A}$.

Our main result is the following.

Theorem 1.1. *Let q be a fixed odd prime power. For every $\varepsilon > 0$, as $g \to \infty$, one has*

$$M(g) = \mathcal{M}_{\mathrm{prin}}(g) + \mathcal{S}(g) + O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big), \tag{1.4}$$

where

$$\mathcal{M}_{\mathrm{prin}}(g) := \frac{\mathscr{P}(1)}{2\zeta_{\mathcal{A}}(2)} q^{2g+2}\left[(2g+2) + \frac{4}{\log q}\frac{\mathscr{P}'}{\mathscr{P}}(1) + 2\zeta_{\mathcal{A}}\big(\tfrac{1}{2}\big)\right] \tag{1.5}$$

and

$$\mathcal{S}(g) := q^{2g/3+2}\,(a_g g + b_g)\,. \tag{1.6}$$

For $m \in \mathbb{Z}$, the coefficients a_m and b_m are real and are given by (4.57), and the sequence $m \mapsto (a_m, b_m)$ has minimal period exactly three.

The principal term $\mathcal{M}_{\mathrm{prin}}(g)$ and the error exponent are those of [3, Theorem 1.4] and [10, Theorem 1.1]. The secondary term is again of order $gq^{2g/3}$, but its coefficients, unlike theirs, depend on g mod 3; Remark 6.2 shows numerically that this is the shape realized by $M(g)$.

Passing to the completed L-function is what makes it possible to treat the even hyperelliptic ensemble along the lines followed for the odd hyperelliptic ensemble in [6] and, for the period-three secondary term, in [11]. It also avoids a computation that the earlier even-degree treatments do not. In [3] and [10] the principal contribution is evaluated by residues at the origin, which brings in the Taylor coefficients of the Euler product there and, in [3], an induction on g carried out in an appendix. Here, as in [6, Section 8], the corresponding contour integral is retained rather than evaluated: it reappears with the opposite sign among the square-dual residues, the two cancel exactly, and no residue at the origin is computed at all. This is a second point at which the present treatment follows the odd-degree argument of [6] rather than the even-degree ones of [3, 10].

The passage to L^* should serve the higher moments as well. Since L^* satisfies the exact two-term central-value formula (2.10), the k-th moment of L^* expands into 2^k blocks of a single shape, indexed by truncation levels $N_1, \dots, N_k \in \{g, g-1\}$, and the factor $(1-q^{-1/2})^k$ returns the k-th moment of $L\big(\frac{1}{2}, \chi_D\big)$; the second and third moments over $\mathcal{H}_{2g+2}$, obtained for $\mathcal{H}_{2g+1}$ in [7], should be within reach in this way. With the four-sum identity of [10, Lemma 2.1] and [3, Lemma 2.4] the same moment expands instead into 4^k blocks of two kinds, and the bookkeeping grows quickly with k (Remark 2.3).

The paper is organized as follows. Section 2 develops the basic function-field notation, the completed L-function and its exact two-term central-value formula, the square-free character-average identity, the normalized Gauss sums, the Poisson summation formula, and the decomposition of $M^*(g)$. Section 3 evaluates the zero-dual contribution. Section 4 analyzes the nonzero square dual variables, proves the principal contour cancellation, evaluates the secondary residues, and establishes the exact period-three structure. Section 5 proves a uniform fixed-V coefficient estimate and uses it to bound the nonsquare contribution. Finally, Section 6 combines the three contributions and completes the proof of Theorem 1.1, and closes with a numerical verification of the period-three secondary term for small q and g (Remark 6.2).

2. Background and setup of the problem

This section collects the material used below: notation and standard facts about $\mathcal{A} = \mathbb{F}_q[x]$ and its quadratic characters, the exact two-term formula for the central value of the completed L-function, the character average over $\mathcal{H}_{2g+2}$, the phase-normalized Gauss sums and the Poisson summation formula, and the decomposition of $M^*(g)$.

2.1. Basic facts about function fields and L-functions.

Throughout the paper, q denotes an odd prime power and $\mathcal{A} = \mathbb{F}_q[x]$. For a nonzero $f \in \mathcal{A}$, we write $d(f) := \deg f$ and define the norm of f by $|f| := q^{d(f)}$. We write $\mathcal{A}^+$ for the set of monic polynomials, $\mathcal{P}$ for the set of monic irreducible polynomials, and $\mathcal{H}$ for the set of monic square-free polynomials in $\mathcal{A}$; in particular $1 \in \mathcal{H}$. Elements of $\mathcal{P}$ are called *prime polynomials*, and the letter P is reserved for them throughout. Unless otherwise stated, a sum over $h \mid f$ is taken over the monic divisors $h \in \mathcal{A}^+$ of f.

In the Vinogradov notation $f \ll g$, equivalently $f = O(g)$, subscripts indicate the parameters on which the implied constant may depend; since q is fixed throughout, it may always depend on q as well, whether or not this is indicated.

For any subset $\mathcal{U}$ of $\mathcal{A}^+$, write $\mathcal{U}_n = \{f \in \mathcal{U} : d(f) = n\}$ and $\mathcal{U}_{\le n} = \{f \in \mathcal{U} : d(f) \le n\}$; both are understood to be empty whenever $n < 0$. The *even hyperelliptic ensemble* of degree $2g+2$ is the family $\mathcal{H}_{2g+2}$.

For $\mathrm{Re}(s) > 1$, the zeta function of $\mathcal{A}$ is given by

$$\zeta_{\mathcal{A}}(s) := \sum_{f \in \mathcal{A}^+} \frac{1}{|f|^s} = \prod_{P \in \mathcal{P}} \left(1 - \frac{1}{|P|^s}\right)^{-1} = \left(1 - q^{1-s}\right)^{-1}$$

(see [12, Chapter 2]); the last expression continues $\zeta_{\mathcal{A}}$ meromorphically to $\mathbb{C}$, and we write $\zeta_{\mathcal{A}}$ also for the continuation, so that $\zeta_{\mathcal{A}}(\frac{1}{2}) = (1 - q^{1/2})^{-1}$. In the variable $u = q^{-s}$ we write $\mathcal{Z}_{\mathcal{A}}(u) := (1 - qu)^{-1}$, so that $\mathcal{Z}_{\mathcal{A}}(q^{-s}) = \zeta_{\mathcal{A}}(s)$ and $\sum_{f \in \mathcal{A}^+} u^{d(f)} = \mathcal{Z}_{\mathcal{A}}(u)$ for $|u| < q^{-1}$; below, $\mathcal{Z}_{\mathcal{A}}$ is also used as a meromorphic function on $\mathbb{C} \setminus \{q^{-1}\}$.

For $a \in \mathcal{A}$ and $P \in \mathcal{P}$, the quadratic residue symbol $\left(\frac{a}{P}\right) \in \{0, \pm 1\}$ is defined by $\left(\frac{a}{P}\right) \equiv a^{(|P|-1)/2} \bmod P$, so that $\left(\frac{a}{P}\right) = 0$ exactly when $P \mid a$. For nonconstant $f \in \mathcal{A}^+$ with prime factorization $f = \prod_P P^{e_P}$, the Jacobi symbol is obtained by multiplicative extension,

$$\left(\frac{a}{f}\right) := \prod_P \left(\frac{a}{P}\right)^{e_P},$$

with the convention $\left(\frac{a}{1}\right) = 1$; thus $\left(\frac{a}{f}\right) = 0$ precisely when $(a, f) \neq 1$. For fixed $f \in \mathcal{A}^+$, the map $a \mapsto \left(\frac{a}{f}\right)$ depends only on a modulo f; we write $\psi_f(a) := \left(\frac{a}{f}\right)$ for $a \in \mathcal{A}$, a quadratic Dirichlet character modulo f, which need not be primitive. Letting f_1 be the *odd part* of f, that is, the unique square-free divisor of f for which f/f_1 is a square, multiplicativity of the Jacobi symbol gives

$$\psi_f(a) = \left(\frac{a}{f_1}\right) \mathbf{1}_{(a,f)=1} \qquad (a \in \mathcal{A}), \tag{2.1}$$

so that ψ_f is induced by the primitive character ψ_{f_1} modulo f_1; in particular ψ_f is principal if and only if f is a square.

Let $\chi_2 : \mathbb{F}_q^\times \to \{\pm 1\}$ be the quadratic character defined by $\chi_2(c) := c^{(q-1)/2}$, regarded as taking values ± 1 in $\mathbb{Z}$. Since q is odd, one has $\left(\frac{c}{P}\right) = \chi_2(c)^{d(P)}$ for $c \in \mathbb{F}_q^\times$ (see [12, Chapter 3]), and therefore

$$\psi_f(c) = \left(\frac{c}{f}\right) = \chi_2(c)^{d(f)} \qquad (c \in \mathbb{F}_q^\times,\ f \in \mathcal{A}^+). \tag{2.2}$$

As usual, a Dirichlet character χ modulo f is called *even* if $\chi(c) = 1$ for every $c \in \mathbb{F}_q^\times$, and *odd* otherwise. By (2.2), ψ_f is even precisely when $d(f)$ is even.

For coprime $A, B \in \mathcal{A}^+$, the quadratic reciprocity law over $\mathcal{A}$ (see [12, Chapter 3]) states that

$$\left(\frac{A}{B}\right) = (-1)^{\frac{q-1}{2} d(A) d(B)} \left(\frac{B}{A}\right).$$

For $D \in \mathcal{H}_{2g+2}$, let $\chi_D := \left(\frac{D}{\cdot}\right)$ be the quadratic character attached to D, as in [3,6,10]. Since $d(D) = 2g+2$ is even, the reciprocity sign above is 1 whenever $(D, f) = 1$, and both symbols vanish otherwise; hence

$$\chi_D(f) = \left(\frac{D}{f}\right) = \left(\frac{f}{D}\right) = \psi_D(f) \qquad (f \in \mathcal{A}^+), \tag{2.3}$$

for every odd prime power q, with no congruence condition such as $q \equiv 1 \bmod 4$. Since D is square-free, its odd part is D itself, so that (2.1) shows χ_D to be the primitive quadratic character modulo D.

The following counting fact will be used repeatedly. Let $h \in \mathcal{A}^+$ and $m \geq d(h)$. Every $H \in \mathcal{A}_m^+$ is uniquely written as $H = A + Bh$ with $d(A) < d(h)$ and $B \in \mathcal{A}_{m-d(h)}^+$; hence each residue class modulo h contains exactly $q^{m-d(h)}$ monic polynomials of degree m, and consequently

$$\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = q^{m-d(h)} \sum_{A \bmod h} \psi_h(A). \tag{2.4}$$

L-functions and the completed L-function. Let χ be a Dirichlet character over $\mathcal{A}$. For $\mathrm{Re}(s) > 1$, the Dirichlet L-function attached to χ is given by the absolutely convergent series and Euler product

$$L(s, \chi) := \sum_{f \in \mathcal{A}^+} \frac{\chi(f)}{|f|^s} = \prod_{P \in \mathcal{P}} \left(1 - \frac{\chi(P)}{|P|^s}\right)^{-1}$$

(see [12, Chapter 4]). In the variable $u = q^{-s}$ one has $|f|^{-s} = u^{d(f)}$, so that, for $|u| < q^{-1}$,

$$\mathcal{L}(u, \chi) := L(s, \chi) = \sum_{n \geq 0} \left(\sum_{f \in \mathcal{A}_n^+} \chi(f) \right) u^n.$$

For $D \in \mathcal{H}_{2g+2}$ the character χ_D is nonprincipal, D being square-free and non-constant, and $\mathcal{L}(u, \chi_D)$ is a polynomial in u of degree at most $2g+1$ (see [12, Proposition 4.3]; cf. [10, Section 2]).

Since $d(D)$ is even, $\mathcal{L}(u,\chi_D)$ has a trivial zero at $u=1$, so that the *completed L-function*

$$\mathcal{L}^*(u,\chi_D) := \frac{\mathcal{L}(u,\chi_D)}{1-u} \tag{2.5}$$

is a polynomial in u of degree exactly $2g$ with $\mathcal{L}^*(0,\chi_D)=1$: it is the numerator of the zeta function of the hyperelliptic curve $y^2=D(x)$, which has genus g over $\mathbb{F}_q$, and it satisfies the functional equation

$$\mathcal{L}^*(u,\chi_D) = (qu^2)^g \mathcal{L}^*\left(\frac{1}{qu},\chi_D\right). \tag{2.6}$$

Since $\mathcal{L}(u,\chi_D)$ is a polynomial in u, the function $s\mapsto \mathcal{L}(q^{-s},\chi_D)$ is entire and continues $L(s,\chi_D)$ to all of $\mathbb{C}$; the central value $L(\frac{1}{2},\chi_D)$ is understood as $\mathcal{L}(q^{-1/2},\chi_D)$. Evaluating (2.5) at $u=q^{-1/2}$ therefore gives $L\big(\frac{1}{2},\chi_D\big)=\mathcal{L}(q^{-1/2},\chi_D)=(1-q^{-1/2})\mathcal{L}^*(q^{-1/2},\chi_D)$. Since $u=q^{-s}$, the function $\mathcal{L}^*(u,\chi_D)$ is the completed L-function $L^*(s,\chi_D)$ of §1; in particular $L^*\big(\frac{1}{2},\chi_D\big)=\mathcal{L}^*(q^{-1/2},\chi_D)$. Summing the identity above over $D\in\mathcal{H}_{2g+2}$ and using the definitions (1.1) and (1.2) yields the recovery relation (1.3).

2.2. **Two-term formula and preliminary lemmas.** We first identify the coefficients of the completed L-function $\mathcal{L}^*(u,\chi_D)$.

Lemma 2.1. *Let $D\in\mathcal{H}_{2g+2}$. Then one has*

$$\mathcal{L}^*(u,\chi_D) = \sum_{f\in\mathcal{A}^+} A_D^*(f)u^{d(f)}, \tag{2.7}$$

where

$$A_D^*(f) := \sum_{\substack{h,m\in\mathcal{A}^+\\ hm=f}} \frac{\chi_D(h)}{|m|} = \frac{1}{|f|}\sum_{h\mid f}|h|\chi_D(h). \tag{2.8}$$

The series in (2.7) converges absolutely for $|u|<q^{-1}$, and (2.7) also holds as an identity of formal power series in u.

Proof. Regrouping the expansion of $\mathcal{L}(u,\chi)$ in §2.1 gives $\mathcal{L}(u,\chi_D)=\sum_{h\in\mathcal{A}^+}\chi_D(h)u^{d(h)}$, absolutely convergent for $|u|<q^{-1}$, while $\sum_{m\in\mathcal{A}^+}u^{d(m)}|m|^{-1}=(1-u)^{-1}$ for $|u|<1$, the coefficient of u^n on the left being $q^{-n}\#\mathcal{A}_n^+=1$. Multiplying $\sum_{h\in\mathcal{A}^+}\chi_D(h)u^{d(h)}$ by $\sum_{m\in\mathcal{A}^+}u^{d(m)}|m|^{-1}$ — legitimate for $|u|<q^{-1}$, where both converge absolutely — and collecting the terms with $hm=f$ gives (2.7) with the first expression in (2.8); substituting $m=f/h$ turns it into the second. The same computation is an identity of formal power series. □

Grouping (2.7) according to degree expresses $\mathcal{L}^*(u,\chi_D)$ as a power series in which the coefficient of u^N is $\sum_{f\in\mathcal{A}_N^+}A_D^*(f)$. Since $\mathcal{L}^*(u,\chi_D)$ is a polynomial of degree $2g$, these coefficients vanish for $N>2g$ – although the individual values $A_D^*(f)$ with $d(f)>2g$ do not – and

$$\mathcal{L}^*(u,\chi_D) = \sum_{N=0}^{2g}\left(\sum_{f\in\mathcal{A}_N^+}A_D^*(f)\right)u^N. \tag{2.9}$$

Lemma 2.2 (Exact two-term formula)**.** *For every $D\in\mathcal{H}_{2g+2}$, one has*

$$\mathcal{L}^*\left(\frac{1}{\sqrt{q}},\chi_D\right) = \sum_{f\in\mathcal{A}^+_{\le g}}\frac{A_D^*(f)}{\sqrt{|f|}} + \sum_{f\in\mathcal{A}^+_{\le g-1}}\frac{A_D^*(f)}{\sqrt{|f|}}. \tag{2.10}$$

Proof. By (2.9), $\mathcal{L}^*(u,\chi_D)$ is a polynomial in u of degree $2g$ whose coefficient of u^N is $\sum_{f\in\mathcal{A}_N^+}A_D^*(f)$ for $0\le N\le 2g$, and by (2.6) it satisfies the same functional equation as $\mathcal{L}(u,\chi_D)$ for $D\in\mathcal{H}_{2g+1}$; the argument of [1, Lemma 3.3] therefore applies verbatim and gives (2.10). □

Remark 2.3 (Comparison with earlier central-value formulas). For $\mathcal{H}_{2g+2}$, the exact central-value identity of [10, Lemma 2.1] and [3, Lemma 2.4] expresses $L(\frac{1}{2},\chi_D)$ as four sums: two Dirichlet-polynomial sums coming from the functional equation, and two correction sums attached to the trivial factor $1-u$. Dividing that factor out first makes (2.6) symmetric and absorbs its effect into the coefficients $A_D^*(f)$ of (2.8), leaving the two sums of a single shape in (2.10).

The square-free character average. The next lemma expresses the average over square-free polynomials in terms of unrestricted polynomial sums. It is [10, Lemma 2.2], the even-degree analogue of [6, Lemma 2.2]; see also [3]. The proof opens by replacing $\chi_D(h)$ with $\psi_h(D)$ through quadratic reciprocity, a step taken in [6] under the hypothesis $q\equiv 1 \bmod 4$; by (2.3) it is available here for every odd prime power q, $d(D)$ being even, and the rest of the argument is unchanged.

Lemma 2.4 (Character average over $\mathcal{H}_{2g+2}$)**.** *For $h \in \mathcal{A}^+$, one has*

$$\sum_{D\in\mathcal{H}_{2g+2}} \chi_D(h) = \sum_{\substack{C\in\mathcal{A}^+_{\le g+1}\\ C\mid h^\infty}} \sum_{H\in\mathcal{A}^+_{2g+2-2d(C)}} \psi_h(H) - q \sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C\mid h^\infty}} \sum_{H\in\mathcal{A}^+_{2g-2d(C)}} \psi_h(H).$$

Here $C \mid h^\infty$ means that every prime polynomial factor of C divides h.

The function-field Perron formula. We will repeatedly use the following coefficient-extraction identities, commonly referred to as the function-field Perron formula; see [3, Lemma 2.8] and [10, Fact 3.1].

Lemma 2.5 (Function-field Perron formula)**.** *Let $a : \mathcal{A}^+ \to \mathbb{C}$, and suppose that $\mathcal{F}(u) := \sum_{f\in\mathcal{A}^+} a(f)u^{d(f)}$ converges absolutely for $|u| < R$, where $R > 0$. If $n \in \mathbb{Z}_{\ge 0}$ and $0 < r < \min\{R, 1\}$, then*

$$\sum_{f\in\mathcal{A}^+_n} a(f) = \frac{1}{2\pi i}\oint_{|u|=r} \frac{\mathcal{F}(u)}{u^{n+1}}\,du, \tag{2.11}$$

$$\sum_{f\in\mathcal{A}^+_{\le n}} a(f) = \frac{1}{2\pi i}\oint_{|u|=r} \frac{\mathcal{F}(u)}{(1-u)u^{n+1}}\,du. \tag{2.12}$$

2.3. **Poisson summation formula.** For a general odd prime power q, the quadratic Gauss sums over $\mathcal{A}$ carry a root-of-unity phase, which is where $q \equiv 1 \bmod 4$ enters [6]. Building that phase into the definition of the Gauss sum makes its local values, its multiplicativity and the Poisson summation formula phase-free for every odd prime power q.

Let $p = \operatorname{char}\mathbb{F}_q$ and let $e_p(\alpha) := \exp\big(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(\alpha)/p\big)$ be the canonical additive character of $\mathbb{F}_q$. With χ_2 as in §2.1, put

$$g_1 := \sum_{\alpha\in\mathbb{F}_q^\times} \chi_2(\alpha)e_p(\alpha), \qquad \varepsilon_q := \frac{g_1}{\sqrt{q}}. \tag{2.13}$$

The classical identity $g_1^2 = \chi_2(-1)q$ (see [9, Chapter 10, Section 3]) gives

$$\varepsilon_q^2 = \chi_2(-1), \qquad \varepsilon_q^4 = 1; \tag{2.14}$$

in particular $|\varepsilon_q| = 1$, so that $\varepsilon_q\overline{\varepsilon_q} = 1$. Thus $\varepsilon_q \in \{\pm 1\}$ if $q \equiv 1 \bmod 4$, and $\varepsilon_q \in \{\pm i\}$ if $q \equiv 3 \bmod 4$; for prime q, Gauss's evaluation gives $\varepsilon_q = 1$ when $q \equiv 1 \bmod 4$ and $\varepsilon_q = i$ when $q \equiv 3 \bmod 4$.

For $a \in \mathbb{F}_q((x^{-1}))$, write $[x^{-1}]a$ for the coefficient of x^{-1} in the Laurent expansion of a, and put $e(a) := e_p\big([x^{-1}]a\big)$. Since $a \mapsto [x^{-1}]a$ is $\mathbb{F}_p$-linear and e_p is an additive character, e is a character of the additive group $\mathbb{F}_q((x^{-1}))$. Every $a \in \mathcal{A}$ has $[x^{-1}]a = 0$, so e is trivial on $\mathcal{A}$.

For $h \in \mathcal{A}^+$, let $\nu(h) \in \{0, 1\}$ be the parity of $d(h)$, so that $\nu(h) \equiv d(h) \bmod 2$.

Definition 2.6 (Phase-normalized Gauss sum)**.** For $h \in \mathcal{A}^+$ and $V \in \mathcal{A}$, define

$$\mathcal{G}(V, \psi_h) := \overline{\varepsilon_q}^{\,\nu(h)} \sum_{u \bmod h} \psi_h(u)e\left(\frac{uV}{h}\right). \tag{2.15}$$

Since ψ_h is a character modulo h and e is trivial on $\mathcal{A}$, the summand in (2.15) depends only on u and V modulo h; the sum is therefore independent of the residue system chosen, and $\mathcal{G}(V, \psi_h)$ depends only on $V \bmod h$.

Lemma 2.7 (Normalized prime Gauss sum)**.** *For every $P \in \mathcal{P}_d$, one has*

$$\mathcal{G}(1, \psi_P) = q^{d/2} = \sqrt{|P|}. \tag{2.16}$$

Proof. Let $\theta_1, \dots, \theta_d$ be the roots of P, put $\theta := \theta_1$ and $K := \mathbb{F}_q(\theta) \cong \mathbb{F}_{q^d}$, so that $f \mapsto f(\theta)$ is a ring isomorphism $\mathcal{A}/(P) \to K$. The roots of P are distinct and lie in K, so for $u \in \mathcal{A}$ with $d(u) < d$ the partial-fraction decomposition of u/P over K reads

$$\frac{u}{P} = \sum_{i=1}^{d} \frac{u(\theta_i)}{P'(\theta_i)} \cdot \frac{1}{x - \theta_i},$$

and $(x - \theta_i)^{-1} = \sum_{n\ge 0} \theta_i^n x^{-n-1}$ gives $[x^{-1}](u/P) = \sum_{i=1}^d u(\theta_i)/P'(\theta_i)$, the d summands being the images of $u(\theta)/P'(\theta)$ under the $\mathbb{F}_q$-automorphisms of K, so that

$$e\Big(\frac{u}{P}\Big) = e_p\left(\operatorname{Tr}_{K/\mathbb{F}_q}\left(\frac{u(\theta)}{P'(\theta)}\right)\right).$$

Let χ_K be the quadratic character of $K^\times$ (with $\chi_K(0) = 0$) and $g_d := \sum_{\bar u\in K} \chi_K(\bar u)e_p\big(\operatorname{Tr}_{K/\mathbb{F}_q}(\bar u)\big)$, so that g_1 is the sum in (2.13). Transporting $\big(\frac{u}{P}\big) \equiv u^{(|P|-1)/2} \bmod P$ along $f \mapsto f(\theta)$ gives $\psi_P(u) = \chi_K(u(\theta))$, and substituting $\bar u \mapsto \bar u P'(\theta)$, which permutes K since $P'(\theta) \ne 0$, gives $\sum_{u \bmod P} \psi_P(u)e(u/P) = \chi_K(P'(\theta))g_d$.

Since $\chi_K = \chi_2 \circ N_{K/\mathbb{F}_q}$ and $N_{K/\mathbb{F}_q}(P'(\theta)) = \prod_{i\neq j}(\theta_i - \theta_j) = (-1)^{d(d-1)/2}\operatorname{disc} P$, the Pellet–Stickelberger theorem [13, Corollary 1] ($\chi_2(\operatorname{disc} P) = (-1)^{d-1}$) gives $\chi_K(P'(\theta)) = \chi_2(-1)^{d(d-1)/2}(-1)^{d-1}$, while the Hasse–Davenport relation [9, Chapter 11, Section 4] gives $g_d = (-1)^{d-1}g_1^d$. The signs $(-1)^{d-1}$ cancel, and $\chi_2(-1) = \varepsilon_q^2$ by (2.14) and $g_1 = \varepsilon_q\sqrt{q}$ by (2.13) give

$$\sum_{u \bmod P} \psi_P(u)e\Big(\frac{u}{P}\Big) = \varepsilon_q^{d(d-1)}\varepsilon_q^d q^{d/2} = \varepsilon_q^{d^2} q^{d/2}. \tag{2.17}$$

Since $d^2 \equiv \nu(P) \bmod 4$ and $\varepsilon_q^4 = 1$, multiplying (2.17) by $\overline{\varepsilon_q}^{\nu(P)}$ as in (2.15) gives (2.16). □

Lemma 2.8 (Phase-free quadratic Gauss sums). (1) *Let $P \in \mathcal{P}_d$, and $i \geq 1$. For $V \in \mathcal{A}\setminus\{0\}$, write $V = P^kV_1$ with $k \geq 0$ and $P \nmid V_1$. Then one has*

$$\mathcal{G}(V, \psi_{P^i}) = \begin{cases} 0 & i \leq k,\ i \text{ odd}, \\ \phi(P^i) & i \leq k,\ i \text{ even}, \\ -|P|^{i-1} & i = k+1,\ i \text{ even}, \\ \big(\frac{V_1}{P}\big)|P|^{i-1}\sqrt{|P|} & i = k+1,\ i \text{ odd}, \\ 0 & i \geq k+2, \end{cases} \tag{2.18}$$

where $\phi(f)$ denotes the polynomial Euler totient function over $\mathcal{A}$.

(2) *For coprime $h_1, h_2 \in \mathcal{A}^+$ and $V \in \mathcal{A}$, one has*

$$\mathcal{G}(V, \psi_{h_1h_2}) = \mathcal{G}(V, \psi_{h_1})\mathcal{G}(V, \psi_{h_2}). \tag{2.19}$$

(3) *For $h \in \mathcal{A}^+$, one has*

$$\mathcal{G}(0, \psi_h) = \phi(h)\mathbf{1}_{h=\square}. \tag{2.20}$$

Proof. We prove (2.18) first, distinguishing the two parities of i. Suppose first that i is even. Then $\nu(P^i) = 0$ and $\psi_{P^i}(u) = \big(\frac{u}{P^i}\big) = \mathbf{1}_{P\nmid u}$. Thus $\mathcal{G}(V, \psi_{P^i}) = \sum_{u \bmod P^i,\, P\nmid u} e\big(uV/P^i\big)$. Removing from the full sum over $u \bmod P^i$ the terms with $P \mid u$, which are parameterized by $u = Pv$ with $v \bmod P^{i-1}$, and using additive orthogonality in the form

$$\sum_{u \bmod P^j} e\Big(\frac{uV}{P^j}\Big) = |P|^j\mathbf{1}_{P^j|V} \qquad (j \geq 0),$$

we obtain $\mathcal{G}(V, \psi_{P^i}) = |P|^i\mathbf{1}_{P^i|V} - |P|^{i-1}\mathbf{1}_{P^{i-1}|V}$. Since $V = P^kV_1$ with $P \nmid V_1$, the indicator $\mathbf{1}_{P^j|V}$ equals 1 for $j \leq k$ and 0 for $j > k$. Applying this to $j = i$ and to $j = i-1$, and using $\phi(P^i) = |P|^i - |P|^{i-1}$, we obtain

$$\mathcal{G}(V, \psi_{P^i}) = \begin{cases} \phi(P^i) & i \leq k, \\ -|P|^{i-1} & i = k+1, \\ 0 & i \geq k+2, \end{cases}$$

which are the entries of (2.18) with i even.

Suppose next that i is odd. Then $\nu(P^i) = \nu(P)$ and $\psi_{P^i}(u) = \big(\frac{u}{P}\big)$. Writing $u = a + Pb$ with $a \bmod P$ and $b \bmod P^{i-1}$, we have $\psi_{P^i}(u) = \big(\frac{a}{P}\big)$ and, since $V = P^kV_1$, $uV/P^i = aV_1/P^{i-k} + bV_1/P^{i-k-1}$; hence (2.15) factors as

$$\mathcal{G}(V, \psi_{P^i}) = \overline{\varepsilon_q}^{\nu(P)}\left(\sum_{a \bmod P}\Big(\frac{a}{P}\Big)e\Big(\frac{aV_1}{P^{i-k}}\Big)\right)\left(\sum_{b \bmod P^{i-1}} e\Big(\frac{bV_1}{P^{i-k-1}}\Big)\right).$$

If $i \geq k+2$, the summand of the second sum depends only on $b \bmod P^{i-k-1}$, e being trivial on $\mathcal{A}$, so that sum equals $|P|^k\sum_{c \bmod P^{i-k-1}} e\big(cV_1/P^{i-k-1}\big) = 0$ by the orthogonality relation above, since $i-k-1 \geq 1$ and $P \nmid V_1$. If $i \leq k+1$, then $bV_1/P^{i-k-1} \in \mathcal{A}$, so the second sum equals $|P|^{i-1}$. For $i \leq k$ one has $aV_1/P^{i-k} \in \mathcal{A}$ as well, and the first sum is $\sum_{a \bmod P}\big(\frac{a}{P}\big) = 0$; for $i = k+1$, replacing a by aV_1^{-1}, where V_1^{-1} is the inverse of V_1 modulo P, and using $\big(\frac{V_1^{-1}}{P}\big) = \big(\frac{V_1}{P}\big)$, turns the first sum into $\big(\frac{V_1}{P}\big)\varepsilon_q^{\nu(P)}\mathcal{G}(1, \psi_P)$, so that $\mathcal{G}(V, \psi_{P^i}) = \big(\frac{V_1}{P}\big)|P|^{i-1}\sqrt{|P|}$ by Lemma 2.7. This proves (2.18).

Now let $(h_1, h_2) = 1$ and put $d_j = d(h_j)$. Writing $u = u_1h_2 + u_2h_1 \bmod h_1h_2$, the Chinese remainder theorem and the quadratic reciprocity law give, as in [6, Lemma 3.2],

$$\sum_{u \bmod h_1h_2} \psi_{h_1h_2}(u)e\Big(\frac{uV}{h_1h_2}\Big) = \chi_2(-1)^{d_1d_2}\prod_{j=1}^{2}\left(\sum_{u_j \bmod h_j}\psi_{h_j}(u_j)e\Big(\frac{u_jV}{h_j}\Big)\right),$$

the sign $\chi_2(-1)^{d_1d_2}$ being -1 when $q \equiv 3 \bmod 4$ and d_1, d_2 are both odd, so that the unnormalized sums are not multiplicative in general. For $j = 1, 2$ one has $\sum_{u_j \bmod h_j}\psi_{h_j}(u_j)e\big(u_jV/h_j\big) = \varepsilon_q^{\nu(h_j)}\mathcal{G}(V, \psi_{h_j})$ by (2.15)

and $|\varepsilon_q| = 1$, and $\chi_2(-1) = \varepsilon_q^2$ by (2.14); hence

$$\mathcal{G}(V, \psi_{h_1 h_2}) = \varepsilon_q^{\nu(h_1)+\nu(h_2)-\nu(h_1h_2)+2d_1d_2}\mathcal{G}(V, \psi_{h_1})\mathcal{G}(V, \psi_{h_2}).$$

Since $\nu(h_1h_2) = \nu(h_1) + \nu(h_2) - 2\nu(h_1)\nu(h_2)$, the exponent equals $2\big(\nu(h_1)\nu(h_2) + d_1d_2\big)$; and $\nu(h_j) \equiv d_j \bmod 2$ gives $\nu(h_1)\nu(h_2) \equiv d_1d_2 \bmod 2$, so the exponent is divisible by 4. As $\varepsilon_q^4 = 1$ by (2.14), this proves (2.19).

Finally, by definition, $\mathcal{G}(0, \psi_h) = \overline{\varepsilon_q}^{\nu(h)} \sum_{u \bmod h} \psi_h(u)$. The character ψ_h is principal precisely when h is a square. In that case $\nu(h) = 0$, and the character sum equals $\phi(h)$; otherwise it vanishes. This proves (2.20). □

With the Gauss sum $\mathcal{G}$ normalized as above, the Poisson summation formula over $\mathcal{A}$ takes the following form for every odd prime power q.

Proposition 2.9 (Poisson summation over $\mathcal{A}$)**.** *Let $h \in \mathcal{A}^+$ and $m \geq 0$. If $d(h)$ is even, then*

$$\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = \frac{q^m}{|h|}\left(\mathcal{G}(0, \psi_h) + (q-1)\sum_{V \in \mathcal{A}^+_{\leq d(h)-m-2}} \mathcal{G}(V, \psi_h) - \sum_{V \in \mathcal{A}^+_{d(h)-m-1}} \mathcal{G}(V, \psi_h)\right). \tag{2.21}$$

If $d(h)$ is odd, then

$$\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = \frac{q^m\sqrt{q}}{|h|} \sum_{V \in \mathcal{A}^+_{d(h)-m-1}} \mathcal{G}(V, \psi_h). \tag{2.22}$$

Proof. First suppose that $m \geq d(h)$. Then $d(h) - m - 2 < 0$ and $d(h) - m - 1 < 0$, so the V-sums in (2.21) and (2.22) are empty, while (2.4) gives $\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = q^{m-d(h)} \sum_{A \bmod h} \psi_h(A)$. If $d(h)$ is even, then $\nu(h) = 0$, so (2.15) gives $\sum_{A \bmod h} \psi_h(A) = \mathcal{G}(0, \psi_h)$, and $|h| = q^{d(h)}$ turns the right-hand side into $q^m\mathcal{G}(0, \psi_h)/|h|$: this is (2.21). If $d(h)$ is odd, then h is not a square, so $\mathcal{G}(0, \psi_h) = 0$ by Lemma 2.8(3) and both sides of (2.22) vanish.

It remains to consider $0 \leq m \leq d(h) - 1$. For $u \bmod h$, set $S_m(u) := \sum_{H \in \mathcal{A}_m^+} e\big(\frac{uH}{h}\big)$. For $B \in \mathcal{A}$, the map $u \mapsto e(uB/h)$ is a character of $\mathcal{A}/h\mathcal{A}$, trivial precisely when $h \mid B$, so that $|h|^{-1} \sum_{u \bmod h} e\big(u(A - H)/h\big) = \mathbf{1}_{A \equiv H \bmod h}$ for $A, H \in \mathcal{A}$. Writing $\psi_h(H) = \sum_{A \bmod h} \psi_h(A)\mathbf{1}_{A \equiv H \bmod h}$ and then making the changes of variables $u \mapsto -u$ and $A \mapsto -A$, we obtain

$$\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = \frac{\psi_h(-1)}{|h|} \sum_{u \bmod h} S_m(u) \sum_{A \bmod h} \psi_h(A) e\left(\frac{uA}{h}\right) = \frac{\psi_h(-1)\varepsilon_q^{\nu(h)}}{|h|} \sum_{u \bmod h} S_m(u)\mathcal{G}(u, \psi_h),$$

the factor $\varepsilon_q^{\nu(h)}$ arising because the inner sum is the *unnormalized* Gauss sum of (2.15) and $\varepsilon_q\overline{\varepsilon_q} = 1$. By (2.2) the scalar factor outside the last sum is $\psi_h(-1)\varepsilon_q^{\nu(h)}/|h| = \chi_2(-1)^{d(h)}\varepsilon_q^{\nu(h)}/|h|$.

Clearly $S_m(0) = q^m$. Let $u \bmod h$ be nonzero, represented by a polynomial with $d(u) < d(h)$, and write $u/h = \sum_{n \geq 1} c_n x^{-n}$. Writing $H = x^m + \sum_{j<m} a_j x^j$, so that H runs over $\mathcal{A}_m^+$ as $a_0, \dots, a_{m-1}$ run independently over $\mathbb{F}_q$, and using $[x^{-1}]\big(ua_jx^j/h\big) = a_jc_{j+1}$, the additivity of e factors $S_m(u)$ into $e(ux^m/h)\prod_{j<m}\sum_{a \in \mathbb{F}_q} e_p(ac_{j+1})$; each inner sum is q or 0 according as c_{j+1} vanishes or not, e_p being nontrivial. Hence, as in [6, Proposition 3.1], $S_m(u) = q^m e(ux^m/h)$ if $c_n = 0$ for every $1 \leq n \leq m$, and $S_m(u) = 0$ otherwise. The first nonvanishing coefficient of u/h is $c_{d(h)-d(u)}$, so the former holds precisely when $d(u) \leq d(h) - m - 1$.

Write each nonzero $u \bmod h$ with $d(u) \leq d(h) - m - 1$ uniquely as $u = cV$ with $c \in \mathbb{F}_q^\times$ and $V \in \mathcal{A}^+$, so that $d(V) = d(u) \leq d(h) - m - 1$. The substitution $u \mapsto c^{-1}u$ in (2.15) together with (2.2) gives $\mathcal{G}(cV, \psi_h) = \chi_2(c)^{d(h)}\mathcal{G}(V, \psi_h)$; and, h and V being monic, $cVx^m/h = cx^{d(V)+m-d(h)} + \cdots$, so that $e(cVx^m/h)$ equals 1 when $d(V) \leq d(h) - m - 2$ and $e_q(c)$ when $d(V) = d(h) - m - 1$. Grouping the nonzero residues accordingly, we obtain

$$\sum_{H \in \mathcal{A}_m^+} \psi_h(H) = \frac{q^m\chi_2(-1)^{d(h)}\varepsilon_q^{\nu(h)}}{|h|}\left[\mathcal{G}(0, \psi_h) + \Sigma_0 \sum_{V \in \mathcal{A}^+_{\leq d(h)-m-2}} \mathcal{G}(V, \psi_h) + \Sigma_1 \sum_{V \in \mathcal{A}^+_{d(h)-m-1}} \mathcal{G}(V, \psi_h)\right], \tag{2.23}$$

where, by (2.2),

$$\Sigma_0 := \sum_{c \in \mathbb{F}_q^\times} \psi_h(c) = \sum_{c \in \mathbb{F}_q^\times} \chi_2(c)^{d(h)}, \qquad \Sigma_1 := \sum_{c \in \mathbb{F}_q^\times} \psi_h(c)e_q(c) = \sum_{c \in \mathbb{F}_q^\times} \chi_2(c)^{d(h)}e_q(c).$$

It remains to evaluate the scalar prefactor and the two constants Σ_0 and Σ_1 in each parity.

Suppose first that $d(h)$ is even. Then $\nu(h) = 0$ and $\chi_2(-1)^{d(h)} = 1$, so the prefactor is $q^m/|h|$. Moreover $\chi_2(c)^{d(h)} = 1$ for every $c \in \mathbb{F}_q^\times$, whence $\Sigma_0 = q - 1$ and $\Sigma_1 = \sum_{c \in \mathbb{F}_q^\times} e_q(c) = -1$, the second because e_q is a nontrivial additive character of $\mathbb{F}_q$. Therefore (2.23) reduces to (2.21).

Suppose now that $d(h)$ is odd. Then h is not a square, so $\mathcal{G}(0, \psi_h) = 0$ by Lemma 2.8(3) and the first term of (2.23) is absent. Here $\chi_2(c)^{d(h)} = \chi_2(c)$, so that $\Sigma_0 = \sum_{c \in \mathbb{F}_q^\times} \chi_2(c) = 0$ and $\Sigma_1 = \sum_{c \in \mathbb{F}_q^\times} \chi_2(c)e_q(c) = g_1 =$

$\varepsilon_q\sqrt{q}$, so that only the term carrying Σ_1 survives in (2.23). Here three powers of ε_q occur: $\psi_h(-1)=\chi_2(-1)=\varepsilon_q^2$ by (2.14), the factor $\varepsilon_q^{\nu(h)}=\varepsilon_q$ left by the normalization, and $\Sigma_1=\varepsilon_q\sqrt{q}$. The prefactor in (2.23) is thus $q^m\varepsilon_q^3/|h|$, and the coefficient of $\sum_{V\in\mathcal{A}^+_{d(h)-m-1}}\mathcal{G}(V,\psi_h)$ is $q^m\varepsilon_q^3\Sigma_1/|h|=q^m\varepsilon_q^4\sqrt{q}/|h|=q^m\sqrt{q}/|h|$, the three exponents summing to 4. Hence (2.23) reduces to (2.22). □

Remark 2.10. Up to the factor $\overline{\varepsilon_q}^{\nu(h)}$, $\mathcal{G}(V,\psi_h)$ is the generalized Gauss sum $G(u,\chi)$ of [6, Section 3], in which the letters u and V play the opposite roles; when $q\equiv 1 \bmod 4$ is prime one has $\varepsilon_q=1$, the two agree, and (2.18) reduces to the table of local values found there. For a general odd prime power q, the unnormalized sum $\sum_{u \bmod P}\psi_P(u)e(u/P)$ carries the phase $\varepsilon_q^{d^2}$ by (2.17), and the factor $\overline{\varepsilon_q}^{\nu(h)}$ built into (2.15) cancels that phase; every value in Lemma 2.8 is free of it, and the three powers of ε_q in the proof of (2.22) combine to $\varepsilon_q^4=1$. Under the hypothesis $q\equiv 1 \bmod 4$ of [6] each factor is separately trivial and no such cancellation is visible. In [6, Section 9] the primality assumption is removed while $q\equiv 1 \bmod 4$ is retained; there the phase $\varepsilon_q\in\{\pm 1\}$ survives in the odd-degree entry of the local table, and the case $\varepsilon_q=-1$ is not carried out.

The exponent must be the parity $\nu(h)$ and not the degree $d(h)$: with $\overline{\varepsilon_q}^{d(h)}$ in place of $\overline{\varepsilon_q}^{\nu(h)}$ in (2.15) it would be $2d_1d_2$ in the proof of (2.19), so that multiplicativity would fail whenever $q\equiv 3 \bmod 4$ and d_1,d_2 are odd, and $3d(h)+1$ in the proof of (2.22), which is not divisible by 4 when $d(h)\equiv 3 \bmod 4$. The same weight is forced in (2.16) by $d^2\equiv\nu(P) \bmod 4$.

2.4. **Setup of the moment problem.** The argument follows the first-moment framework of [3, Section 3] and [10, Section 3], the new feature being the divisor weight carried by the coefficients $A_D^*(f)$ of (2.8).

For $N\in\{g,g-1\}$, define

$$\mathcal{S}_N^*:=\sum_{D\in\mathcal{H}_{2g+2}}\sum_{f\in\mathcal{A}^+_{\le N}}\frac{A_D^*(f)}{\sqrt{|f|}}. \tag{2.24}$$

Substituting (2.8) into (2.24) and interchanging the finite sums, we obtain

$$\mathcal{S}_N^*=\sum_{f\in\mathcal{A}^+_{\le N}}\frac{1}{|f|^{3/2}}\sum_{h\mid f}|h|\sum_{D\in\mathcal{H}_{2g+2}}\chi_D(h). \tag{2.25}$$

Applying the identity in Lemma 2.4 to the innermost character average in (2.25), we obtain

$$\mathcal{S}_N^*=\sum_{f\in\mathcal{A}^+_{\le N}}\frac{1}{|f|^{3/2}}\sum_{h\mid f}|h|\Bigg(\sum_{\substack{C\in\mathcal{A}^+_{\le g+1}\\ C\mid h^\infty}}\sum_{H\in\mathcal{A}^+_{2g-2d(C)+2}}\psi_h(H)-q\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C\mid h^\infty}}\sum_{H\in\mathcal{A}^+_{2g-2d(C)}}\psi_h(H)\Bigg). \tag{2.26}$$

As in [6, Section 4] (see also [10, (3.2)] and [3, Section 3]), we first discard the terms with $C\in\mathcal{A}^+_{g+1}$ in the first sum of (2.26). There $2g+2-2d(C)=0$, so the inner sum over H equals $\psi_h(1)=1$, while $\#\{C\in\mathcal{A}^+_{g+1}: C\mid h^\infty\}\ll_\varepsilon q^{\varepsilon g}$ uniformly for $d(h)\le g$. Writing $f=hr$, these terms therefore contribute

$$\ll_\varepsilon q^{\varepsilon g}\sum_{\substack{h,r\in\mathcal{A}^+\\ d(h)+d(r)\le N}}|h|^{-1/2}|r|^{-3/2}\ll_\varepsilon q^{\varepsilon g}\sum_{a+b\le N}q^{(a-b)/2}\ll_\varepsilon q^{N/2+\varepsilon g}.$$

Consequently,

$$\mathcal{S}_N^*=\sum_{f\in\mathcal{A}^+_{\le N}}\frac{1}{|f|^{3/2}}\sum_{h\mid f}|h|\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C\mid h^\infty}}\Bigg(\sum_{H\in\mathcal{A}^+_{2g-2d(C)+2}}\psi_h(H)-q\sum_{H\in\mathcal{A}^+_{2g-2d(C)}}\psi_h(H)\Bigg)+O_\varepsilon\Big(q^{N/2+\varepsilon g}\Big). \tag{2.27}$$

Let $\mathcal{S}_{N,\mathrm{e}}^*$ and $\mathcal{S}_{N,\mathrm{o}}^*$ denote the contributions to the sum in (2.27) from divisors $h\mid f$ of even and odd degree, respectively. Then

$$\mathcal{S}_N^*=\mathcal{S}_{N,\mathrm{e}}^*+\mathcal{S}_{N,\mathrm{o}}^*+O_\varepsilon\Big(q^{N/2+\varepsilon g}\Big). \tag{2.28}$$

For $h\in\mathcal{A}^+$ and $C\mid h^\infty$ put

$$\lambda=\lambda(h,C):=d(h)-2g+2d(C), \tag{2.29}$$

and define

$$s_{\mathrm{e}}(h;C) := (q-1)\sum_{V\in\mathcal{A}^+_{\le\lambda-4}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} - \sum_{V\in\mathcal{A}^+_{\lambda-3}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} - \frac{q-1}{q}\sum_{V\in\mathcal{A}^+_{\le\lambda-2}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} + \frac{1}{q}\sum_{V\in\mathcal{A}^+_{\lambda-1}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}}, \tag{2.30}$$

$$s_{\mathrm{o}}(h;C) := \sum_{V\in\mathcal{A}^+_{\lambda-3}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} - \frac{1}{q}\sum_{V\in\mathcal{A}^+_{\lambda-1}}\frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}}. \tag{2.31}$$

We first consider the even-degree contribution. Apply (2.21) to the two sums over H in $\mathcal{S}^*_{N,\mathrm{e}}$: the factor $|h|$ in (2.27) cancels the denominator $|h|$ in (2.21), and with $q^{2g+2-2d(C)} = q^{2g+2}/|C|^2$, $q^{2g+1-2d(C)} = q^{2g+1}/|C|^2$ and $1-q^{-1} = \zeta_{\mathcal{A}}(2)^{-1}$ the terms carrying a nonzero dual variable V assemble into $s_{\mathrm{e}}(h;C)$. Writing $f = hm$ with $m\in\mathcal{A}^+$, so that $d(f)\le N$ becomes $d(h)+d(m)\le N$, we obtain

$$\mathcal{S}^*_{N,\mathrm{e}} = \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)}\sum_{\substack{h,m\in\mathcal{A}^+\\ d(h)+d(m)\le N\\ d(h)\text{ even}}}\frac{\mathcal{G}(0,\psi_h)}{|h|^{3/2}|m|^{3/2}}\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C|h^\infty}}\frac{1}{|C|^2} + q^{2g+2}\sum_{\substack{h,m\in\mathcal{A}^+\\ d(h)+d(m)\le N\\ d(h)\text{ even}}}\frac{1}{|h|\,|m|^{3/2}}\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C|h^\infty}}\frac{s_{\mathrm{e}}(h;C)}{|C|^2}. \tag{2.32}$$

Split $s_{\mathrm{e}}(h;C) = s_{\mathrm{e}}(h;C)_{\square} + s_{\mathrm{e}}(h;C)_{\neq\square}$ according as the dual variable V is a nonzero square or a nonsquare. Since $d(h)$ is even, the two exact-degree sums in (2.30) have odd degree and so contain no squares, so that $s_{\mathrm{e}}(h;C)_{\square}$ comes from the two cumulative sums alone. Accordingly

$$\mathcal{S}^*_{N,\mathrm{e}} = \mathcal{M}^*_N + \mathcal{S}^*_{N,\mathrm{e}}(V=\square) + \mathcal{S}^*_{N,\mathrm{e}}(V\neq\square), \tag{2.33}$$

where $\mathcal{M}^*_N$ is the first sum in (2.32), while $\mathcal{S}^*_{N,\mathrm{e}}(V=\square)$ and $\mathcal{S}^*_{N,\mathrm{e}}(V\neq\square)$ are the second sum with $s_{\mathrm{e}}(h;C)$ replaced by $s_{\mathrm{e}}(h;C)_{\square}$ and by $s_{\mathrm{e}}(h;C)_{\neq\square}$, respectively.

We next consider the odd-degree contribution. Applying (2.22) to the two sums over H in $\mathcal{S}^*_{N,\mathrm{o}}$ in the same way, the extra factor $\sqrt{q}$ raising the prefactor to $q^{2g+5/2}$, we obtain

$$\mathcal{S}^*_{N,\mathrm{o}} = q^{2g+5/2}\sum_{\substack{h,m\in\mathcal{A}^+\\ d(h)+d(m)\le N\\ d(h)\text{ odd}}}\frac{1}{|h|\,|m|^{3/2}}\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C|h^\infty}}\frac{s_{\mathrm{o}}(h;C)}{|C|^2}. \tag{2.34}$$

Here $d(h)$ is odd, so both degrees in (2.31) are even and square dual variables occur in both sums. Splitting $s_{\mathrm{o}}(h;C) = s_{\mathrm{o}}(h;C)_{\square} + s_{\mathrm{o}}(h;C)_{\neq\square}$ as before gives

$$\mathcal{S}^*_{N,\mathrm{o}} = \mathcal{S}^*_{N,\mathrm{o}}(V=\square) + \mathcal{S}^*_{N,\mathrm{o}}(V\neq\square), \tag{2.35}$$

the two terms being (2.34) with $s_{\mathrm{o}}(h;C)$ replaced by $s_{\mathrm{o}}(h;C)_{\square}$ and by $s_{\mathrm{o}}(h;C)_{\neq\square}$, respectively.

Finally, Lemma 2.2 and (2.24) give $M^*(g) = \sum_{N\in\{g,g-1\}}\mathcal{S}^*_N$. Summing (2.28) over $N\in\{g,g-1\}$ and using (2.33) and (2.35), we obtain, after renaming ε,

$$M^*(g) = \mathcal{M}^* + \mathcal{S}^*(V=\square) + \mathcal{S}^*(V\neq\square) + O_\varepsilon\left(q^{g(1+\varepsilon)/2}\right), \tag{2.36}$$

where

$$\mathcal{M}^* := \sum_{N\in\{g,g-1\}}\mathcal{M}^*_N, \tag{2.37}$$

$$\mathcal{S}^*(V=\square) := \sum_{N\in\{g,g-1\}}\left(\mathcal{S}^*_{N,\mathrm{e}}(V=\square) + \mathcal{S}^*_{N,\mathrm{o}}(V=\square)\right), \tag{2.38}$$

$$\mathcal{S}^*(V\neq\square) := \sum_{N\in\{g,g-1\}}\left(\mathcal{S}^*_{N,\mathrm{e}}(V\neq\square) + \mathcal{S}^*_{N,\mathrm{o}}(V\neq\square)\right). \tag{2.39}$$

3. Contribution from the zero dual variable

The purpose of this section is to evaluate the zero-dual contribution $\mathcal{M}^*$ of (2.36), which produces the principal term of Theorem 1.1.

Put

$$\mathcal{C}^*(v) := \prod_{P\in\mathcal{P}}\left(1-\frac{v^{d(P)}}{|P|(|P|+1)}\right). \tag{3.1}$$

Since $q^{d(P)} = |P|$, this is the Euler product $\mathcal{C}$ of [6, (5.4)] and [3, (4.2)] in the rescaled variable:

$$\mathcal{C}^*(v) = \mathcal{C}(v/q), \qquad \text{where} \quad \mathcal{C}(u) = \prod_{P\in\mathcal{P}} \left(1 - \frac{u^{d(P)}}{|P|+1}\right) \tag{3.2}$$

(in [10] both $\mathcal{C}$ and $\mathscr{P}$ are denoted C, the two being the same product in the variables u and s). The rescaling is chosen so that $\mathcal{C}^*$ is the Euler product $\mathscr{P}$ of §1 in the variable $v = q^{1-s}$:

$$\mathscr{P}(s) = \mathcal{C}^*\big(q^{1-s}\big), \qquad \mathcal{C}^*(1) = \mathscr{P}(1). \tag{3.3}$$

By (3.2) the properties of $\mathcal{C}$ recorded in [6, (5.4)–(5.5)] transfer at once: the product $\mathcal{C}^*$ converges absolutely for $|v| < q$, and there

$$\mathcal{C}^*(v) = \left(1 - \frac{v}{q}\right)\mathcal{K}(v), \qquad \text{where} \quad \mathcal{K}(v) := \prod_{P\in\mathcal{P}} \left(1 + \frac{(v/q)^{d(P)}}{(1+|P|)\big(|P| - (v/q)^{d(P)}\big)}\right). \tag{3.4}$$

The product $\mathcal{K}$ converges absolutely for $|v| < q^2$, so $(1 - v/q)\mathcal{K}(v)$ continues $\mathcal{C}^*$ holomorphically to $|v| < q^2$; from here on $\mathcal{C}^*$ denotes this continuation. Equivalently,

$$\frac{\mathcal{C}^*(v)}{q - v} = \frac{\mathcal{K}(v)}{q} \qquad (|v| < q^2), \tag{3.5}$$

the apparent singularity on the left at $v = q$ being removable.

Proposition 3.1. *For every $\varepsilon > 0$ and every $1 < \rho < q$, one has*

$$\mathcal{M}^* = \mathcal{Q}_g + \mathcal{R}_0(g;\rho) + O_\varepsilon(q^{\varepsilon g}), \tag{3.6}$$

where

$$\mathcal{Q}_g := \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \frac{\mathscr{P}(1)}{2(1-q^{-1/2})} \left[(2g+2) + \frac{4}{\log q}\frac{\mathscr{P}'}{\mathscr{P}}(1) + 2\zeta_{\mathcal{A}}\big(\tfrac{1}{2}\big)\right] \tag{3.7}$$

and

$$\mathcal{R}_0(g;\rho) := \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \sum_{N\in\{g,g-1\}} \frac{1}{2\pi i} \oint_{|u|=\rho} \frac{\mathcal{C}^*(u^2)\,du}{(1-q^{-1/2}u)(1-u)^2(1+u)\,u^{N+1}}. \tag{3.8}$$

3.1. **The principal generating function.** For $|u| < 1$, define

$$\mathcal{A}(u) := \sum_{l,m\in\mathcal{A}^+} \frac{\phi(l^2)}{|l|^3|m|^{3/2}} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1} u^{2d(l)+d(m)}. \tag{3.9}$$

Since $2d(l) + d(m) = d(l^2m)$, grouping the terms of (3.9) according to $f = l^2m$ gives $\mathcal{A}(u) = \sum_{f\in\mathcal{A}^+} a(f)u^{d(f)}$, where

$$a(f) := \sum_{\substack{l,m\in\mathcal{A}^+ \\ l^2m=f}} \frac{\phi(l^2)}{|l|^3|m|^{3/2}} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1}.$$

We now evaluate $\mathcal{A}(u)$ in terms of the Euler product $\mathcal{C}^*$.

Lemma 3.2. *The series $\mathcal{A}(u)$ converges absolutely for $|u| < 1$, and there*

$$\mathcal{A}(u) = \frac{\mathcal{C}^*(u^2)}{(1-q^{-1/2}u)(1-u^2)}. \tag{3.10}$$

Moreover $\mathcal{K}$ is nonvanishing on $|v| < q^2$, so that the only zero of $\mathcal{C}^*$ on $|v| < q^2$ is the simple zero at $v = q$. Consequently

$$\mathcal{A}(u) = \frac{(1+q^{-1/2}u)\mathcal{K}(u^2)}{1-u^2}, \tag{3.11}$$

and $\mathcal{A}$ continues meromorphically to $|u| < q$, its only poles there being simple poles at $u = \pm 1$.

Proof. Since $\phi(l^2) \le |l|^2$ and $\prod_{P|l}(1 - |P|^{-2})^{-1} \le \zeta_{\mathcal{A}}(2)$, the terms of (3.9) are dominated in modulus by $\zeta_{\mathcal{A}}(2)|l|^{-1}|m|^{-3/2}|u|^{2d(l)+d(m)}$, whose sum over $l, m \in \mathcal{A}^+$ is finite for $|u| < 1$. Hence $\mathcal{A}(u)$ converges absolutely for $|u| < 1$ and the double sum factors as

$$\mathcal{A}(u) = \left(\sum_{l\in\mathcal{A}^+} \frac{\phi(l^2)}{|l|^3} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1} u^{2d(l)}\right)\left(\sum_{m\in\mathcal{A}^+} \frac{u^{d(m)}}{|m|^{3/2}}\right).$$

The m-factor is $(1 - q^{-1/2}u)^{-1}$; for the l-factor, the identity

$$\frac{\phi(l^2)}{|l|^3} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1} = \frac{1}{|l|} \prod_{P|l} \frac{|P|}{|P|+1},$$

in which both sides are multiplicative and agree at $l = P^j$, together with $u^{2d(l)}/|l| = (u^2/q)^{d(l)}$, gives

$$\sum_{l\in\mathcal{A}^+} \frac{\phi(l^2)}{|l|^3} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1} u^{2d(l)} = \sum_{l\in\mathcal{A}^+} \left(\frac{u^2}{q}\right)^{d(l)} \prod_{P|l} \frac{|P|}{|P|+1}$$
$$= \mathcal{C}\left(\frac{u^2}{q}\right) \mathcal{Z}_{\mathcal{A}}\left(\frac{u^2}{q}\right) = \frac{\mathcal{C}^*(u^2)}{1-u^2}.$$

The second equality is [6, Section 5] (see also [10, (3.15)] and [3, (4.10)]); the third holds by (3.2) and $\mathcal{Z}_{\mathcal{A}}(u^2/q) = (1-u^2)^{-1}$. This proves (3.10).

The nonvanishing of $\mathcal{K}$ on $|v| < q^2$ follows from (3.4): no local factor vanishes there, since $\left|(v/q)^{d(P)}\right| < |P|$, and the product converges absolutely. Substituting (3.4) into (3.10) gives $\mathcal{A}(u) = (1+q^{-1/2}u)\mathcal{K}(u^2)(1-u^2)^{-1}$ for $|u| < 1$, and the right-hand side is meromorphic on $|u| < q$ with only the simple poles $u = \pm 1$. This proves (3.11). □

3.2. **Proof of Proposition 3.1.** By (2.37), $\mathcal{M}^* = \sum_{N\in\{g,g-1\}} \mathcal{M}_N^*$. Note first that $\mathcal{R}_0(g;\rho)$ is well defined for every $1 < \rho < q$: by (3.4) each of the two integrands in (3.8) satisfies

$$\frac{\mathcal{C}^*(u^2)}{(1-q^{-1/2}u)(1-u)^2(1+u)u^{N+1}} = \frac{(1+q^{-1/2}u)\mathcal{K}(u^2)}{(1-u)^2(1+u)u^{N+1}} \qquad (N \in \{g, g-1\}), \tag{3.12}$$

and the right-hand side is holomorphic on $0 < |u| < q$ apart from the poles at $u = \pm 1$, the apparent singularity at $u = q^{1/2}$ being removable.

Proof. By Lemma 2.8(3) one has $\mathcal{G}(0, \psi_h) = \phi(h)\mathbf{1}_{h=\square}$; in particular the zero dual variable occurs only for square h, so that $d(h)$ is even, $\nu(h) = 0$, and the phase normalization of §2.3 plays no role in this section. Therefore

$$\mathcal{M}_N^* = \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \sum_{\substack{h,m\in\mathcal{A}^+ \\ d(h)+d(m)\le N \\ h=\square}} \frac{\phi(h)}{|h|^{3/2}|m|^{3/2}} \sum_{\substack{C\in\mathcal{A}^+_{\le g} \\ C|h^\infty}} \frac{1}{|C|^2}.$$

As in [6, Section 5], the inner C-sum may be extended to all $C \mid h^\infty$: uniformly for $d(h) \le g$,

$$\sum_{\substack{C\in\mathcal{A}^+_{\le g} \\ C|h^\infty}} \frac{1}{|C|^2} = \prod_{P|h} \left(1 - \frac{1}{|P|^2}\right)^{-1} + O_\varepsilon\left(q^{-(2-\varepsilon)g}\right),$$

and the resulting error contributes $O_\varepsilon(q^{\varepsilon g})$. Writing $h = l^2$, this gives

$$\mathcal{M}_N^* = \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \sum_{\substack{l,m\in\mathcal{A}^+ \\ 2d(l)+d(m)\le N}} \frac{\phi(l^2)}{|l|^3|m|^{3/2}} \prod_{P|l} \left(1 - \frac{1}{|P|^2}\right)^{-1} + O_\varepsilon(q^{\varepsilon g}). \tag{3.13}$$

Since $\mathcal{A}(u)$ converges absolutely for $|u| < 1$, the cumulative Perron formula (2.12), applied to the main sum in (3.13) with $0 < r < 1$, gives

$$\mathcal{M}_N^* = \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \frac{1}{2\pi i} \oint_{|u|=r} \frac{\mathcal{A}(u)}{(1-u)u^{N+1}}\, du + O_\varepsilon(q^{\varepsilon g}). \tag{3.14}$$

Fix $1 < \rho < q$. By (3.11) the integrand of (3.14) is (3.12); since $\mathcal{K}(u^2)$ is holomorphic and nonzero for $|u| < q$, it has a double pole at $u = 1$, a simple pole at $u = -1$ and no other pole in $r < |u| < \rho$. Shifting the contour from $|u| = r$ to $|u| = \rho$, we obtain

$$\mathcal{M}_N^* = \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \frac{1}{2\pi i} \oint_{|u|=\rho} \frac{\mathcal{A}(u)}{(1-u)u^{N+1}}\, du - \frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)} \left(\mathrm{Res}_{u=1} + \mathrm{Res}_{u=-1}\right) \frac{\mathcal{A}(u)}{(1-u)u^{N+1}} + O_\varepsilon(q^{\varepsilon g}). \tag{3.15}$$

For the residue calculations we use (3.10), which holds on $0 < |u| < q$ by meromorphic continuation:

$$\frac{\mathcal{A}(u)}{(1-u)u^{N+1}} = \frac{\mathcal{C}^*(u^2)u^{-N-1}}{(1-q^{-1/2}u)(1-u)^2(1+u)},$$

with a double pole at $u = 1$ and a simple pole at $u = -1$. Writing $G(u) := \mathcal{C}^*(u^2)u^{-N-1}\big((1-q^{-1/2}u)(1+u)\big)^{-1}$, so that $\mathcal{A}(u)\big((1-u)u^{N+1}\big)^{-1} = G(u)(1-u)^{-2}$, one has

$$\mathrm{Res}_{u=1} \frac{\mathcal{A}(u)}{(1-u)u^{N+1}} = G'(1),$$

and logarithmic differentiation of G at $u=1$ gives

$$-\operatorname{Res}_{u=1}\frac{\mathcal{A}(u)}{(1-u)u^{N+1}}=\frac{\mathcal{C}^*(1)}{2(1-q^{-1/2})}\left[(N+1)+\frac{1}{2}-\frac{q^{-1/2}}{1-q^{-1/2}}-2\frac{(\mathcal{C}^*)'(1)}{\mathcal{C}^*(1)}\right],$$

while at $u=-1$

$$-\operatorname{Res}_{u=-1}\frac{\mathcal{A}(u)}{(1-u)u^{N+1}}=\frac{(-1)^N\mathcal{C}^*(1)}{4(1+q^{-1/2})}.$$

Substituting these residue calculations into (3.15), we obtain

$$\mathcal{M}_N^*=\frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)}\Bigg\{\frac{\mathcal{C}^*(1)}{2(1-q^{-1/2})}\left[(N+1)+\frac{1}{2}-\frac{q^{-1/2}}{1-q^{-1/2}}-2\frac{(\mathcal{C}^*)'(1)}{\mathcal{C}^*(1)}\right]$$
$$+\frac{(-1)^N\mathcal{C}^*(1)}{4(1+q^{-1/2})}+\frac{1}{2\pi i}\oint_{|u|=\rho}\frac{\mathcal{A}(u)}{(1-u)u^{N+1}}\,du\Bigg\}+O_\varepsilon(q^{\varepsilon g}). \tag{3.16}$$

Summing (3.16) over $N\in\{g,g-1\}$ and using (2.37), the contributions from the residue at $u=-1$ cancel, since $(-1)^g+(-1)^{g-1}=0$. Moreover, $\sum_{N\in\{g,g-1\}}(N+1)=2g+1$. Thus the total contribution of the residues is

$$\frac{\mathcal{C}^*(1)}{2(1-q^{-1/2})}\left[(2g+2)-\frac{2q^{-1/2}}{1-q^{-1/2}}-\frac{4(\mathcal{C}^*)'(1)}{\mathcal{C}^*(1)}\right].$$

By (3.3) one has $\mathcal{C}^*(1)=\mathscr{P}(1)$, and logarithmic differentiation at $s=1$ gives $\frac{\mathscr{P}'}{\mathscr{P}}(1)=-(\log q)\frac{(\mathcal{C}^*)'}{\mathcal{C}^*}(1)$. Moreover $-\frac{q^{-1/2}}{1-q^{-1/2}}=\frac{1}{1-q^{1/2}}=\zeta_{\mathcal{A}}\big(\frac{1}{2}\big)$. It follows from (3.7) that the total residue contribution, multiplied by $q^{2g+2}/\zeta_{\mathcal{A}}(2)$, equals $\mathcal{Q}_g$. Moreover, by (3.10) the contour terms in the sum of (3.16) add up to $\mathcal{R}_0(g;\rho)$. This proves (3.6). □

We retain the displaced contour $\mathcal{R}_0(g;\rho)$ explicitly, rather than estimating it, because the principal square-dual contribution produces $-\mathcal{R}_0(g;\rho)$ and the two cancel exactly in §6.

4. Contribution from square dual variables

The purpose of this section is to evaluate the square-dual contribution $\mathcal{S}^*(V=\square)$ of (2.36). Recall from (2.38) that

$$\mathcal{S}^*(V=\square)=\sum_{N\in\{g,g-1\}}\big(\mathcal{S}^*_{N,\mathrm{e}}(V=\square)+\mathcal{S}^*_{N,\mathrm{o}}(V=\square)\big). \tag{4.1}$$

Proposition 4.1. *For every $\varepsilon>0$ and every $1<\rho<q$, one has*

$$\mathcal{S}^*(V=\square)=-\mathcal{R}_0(g;\rho)+\mathcal{S}_0(g)+O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big), \tag{4.2}$$

where $\mathcal{R}_0(g;\rho)$ is as in (3.8) and $\mathcal{S}_0(g)$ as in (4.39), evaluated explicitly in Corollary 4.11.

4.1. **The generating series $\mathcal{B}$.** We first record the Euler-product input that governs both parity sectors. For $h\in\mathcal{A}^+$, define

$$A_h(z):=\sum_{Y\in\mathcal{A}^+}z^{d(Y)}\frac{\mathcal{G}(Y^2,\psi_h)}{\sqrt{|h|}}. \tag{4.3}$$

This is the phase-normalized counterpart of the function $A_f(z)$ of [10, Section 3.2].

Lemma 4.2. *For every $h\in\mathcal{A}^+$, the series $A_h(z)$ converges absolutely for $|z|<q^{-1}$, and*

$$\big|A_h(z)\big|\le\frac{1}{1-q|z|}\qquad(|z|<q^{-1}), \tag{4.4}$$

uniformly in h.

Proof. For $h=\prod_{P|h}P^{i_P}\in\mathcal{A}^+$, put $\mathrm{sq}(h):=\prod_{P|h}P^{\lfloor i_P/2\rfloor}$. We shall use two auxiliary facts: $\mathcal{G}(Y^2,\psi_h)$ vanishes unless $\mathrm{sq}(h)\mid Y$, and $\big|\mathcal{G}(Y^2,\psi_h)\big|\le\sqrt{|h|}\,\big|\mathrm{sq}(h)\big|$. Since $\mathcal{G}(Y^2,\psi_h)$ is multiplicative in h by (2.19), as are $|h|$ and $\mathrm{sq}(h)$, it suffices to prove both facts for $h=P^i$ with $i\ge1$.

So let $h=P^i$ and write $Y=P^vY_0$ with $v\ge0$ and $P\nmid Y_0$, so that $Y^2=P^{2v}Y_0^2$. Applying (2.18) with $k=2v$, the value $\mathcal{G}(Y^2,\psi_{P^i})$ vanishes unless $i\le2v$ with i even, or $i=2v+1$. In both surviving cases $i\le2v+1$, that is, $v\ge\lfloor i/2\rfloor$, which says exactly that $\mathrm{sq}(P^i)\mid Y$; this is the first fact for $h=P^i$. In those same cases

$$\frac{\big|\mathcal{G}(Y^2,\psi_{P^i})\big|}{|P|^{i/2}}=\begin{cases}\phi(P^i)|P|^{-i/2}<|P|^{i/2}=|P|^{\lfloor i/2\rfloor}, & i\le2v,\ i\text{ even},\\ |P|^{i-1/2}|P|^{-i/2}=|P|^{(i-1)/2}=|P|^{\lfloor i/2\rfloor}, & i=2v+1,\end{cases}$$

which is the second fact for $h = P^i$.

Now let $M \geq 0$. The number of $Y \in \mathcal{A}_M^+$ divisible by $\mathrm{sq}(h)$ is $q^{M-d(\mathrm{sq}(h))}$ if $d(\mathrm{sq}(h)) \leq M$, and zero otherwise, so the above two auxiliary facts give

$$\sum_{Y \in \mathcal{A}_M^+} \frac{|\mathcal{G}(Y^2, \psi_h)|}{\sqrt{|h|}} \leq |\mathrm{sq}(h)| q^{M-d(\mathrm{sq}(h))} = q^M.$$

Grouping (4.3) according to $d(Y)$ therefore bounds $|A_h(z)|$ by $\sum_{M \geq 0} q^M |z|^M = (1 - q|z|)^{-1}$, which is finite for $|z| < q^{-1}$; this proves both assertions. □

We also introduce the two-variable generating series that encodes the h-sum after the C-extension. We denote its second variable by w' to distinguish it from the variable w in which the Perron formula (2.12) is applied below; the two variables will later be related by $w' = w/(qz^{1/2})$. For $q^{-2} < |z| < q^{-1}$ and $|w'|$ sufficiently small, define

$$\mathcal{B}(z, w') := \sum_{h \in \mathcal{A}^+} (w')^{d(h)} A_h(z) \prod_{P | h} \left(1 - \frac{1}{|P|^2 z^{d(P)}}\right)^{-1}. \tag{4.5}$$

The series $\mathcal{B}(z, w')$ is the phase-normalized counterpart of the series of [6, Lemma 6.2], and the following lemma is the corresponding form of [6, Lemmas 6.2–6.3]. Its proof follows those lemmas, and indicates where the phase normalization enters.

Lemma 4.3 (Euler-product factorization of $\mathcal{B}$). *For $q^{-2} < |z| < q^{-1}$ and $|w'|$ sufficiently small, the series $\mathcal{B}(z, w')$ converges absolutely, and it admits the following factorization.*

(1) *One has*

$$\mathcal{B}(z, w') = \mathcal{Z}_{\mathcal{A}}(z) \mathcal{Z}_{\mathcal{A}}(w') \mathcal{Z}_{\mathcal{A}}(q(w')^2 z) \prod_{P \in \mathcal{P}} \mathcal{B}_P(z, w'), \tag{4.6}$$

where, writing $Z_P := z^{d(P)}$ and $W_P := (w')^{d(P)}$ for $P \in \mathcal{P}$, the local factor is

$$\mathcal{B}_P(z, w') = 1 + \frac{W_P\big(1 - |P|^2 Z_P^2\big) - |P|(|P| - 1) Z_P W_P^2 - |P| Z_P \big(1 - |P| Z_P\big) W_P^3}{|P|^2 Z_P - 1}. \tag{4.7}$$

Moreover, the Euler product $\prod_{P \in \mathcal{P}} \mathcal{B}_P(z, w')$ converges absolutely under the weaker restriction

$$|w'| < q|z|, \qquad |w'| < q^{-1/2}, \qquad |w'z| < q^{-1}. \tag{4.8}$$

(2) *For each $P \in \mathcal{P}$, define*

$$\mathcal{D}_P(z, w') := \mathcal{B}_P(z, w') \left(1 - \frac{W_P}{|P|^2 Z_P}\right) (1 - W_P^2)^{-1}. \tag{4.9}$$

Then one has

$$\mathcal{B}(z, w') = \mathcal{Z}_{\mathcal{A}}(z) \mathcal{Z}_{\mathcal{A}}(w') \mathcal{Z}_{\mathcal{A}}(q(w')^2 z) \mathcal{Z}_{\mathcal{A}}\left(\frac{w'}{q^2 z}\right) \mathcal{Z}_{\mathcal{A}}((w')^2)^{-1} \prod_{P \in \mathcal{P}} \mathcal{D}_P(z, w'). \tag{4.10}$$

The right-hand side furnishes a meromorphic continuation of $\mathcal{B}(z, w')$ to the region

$$|z| > q^{-2}, \qquad |w'|^2 < q|z|, \qquad |w'| < q^3 |z|^2, \qquad |w'| < 1, \qquad |w'z| < q^{-1}, \tag{4.11}$$

where the Euler product $\prod_{P \in \mathcal{P}} \mathcal{D}_P(z, w')$ converges absolutely and uniformly on compact subsets.

Proof. We first show that $\mathcal{B}(z, w')$ converges absolutely for $q^{-2} < |z| < q^{-1}$ and $|w'|$ sufficiently small, which justifies the rearrangement into an Euler product made below. By (4.4) one has $|A_h(z)| \leq (1 - q|z|)^{-1}$ for $|z| < q^{-1}$, uniformly in h, while $|z| > q^{-2}$ gives $\prod_{P|h} \big|1 - |P|^{-2} z^{-d(P)}\big|^{-1} \leq \big(1 - q^{-2}/|z|\big)^{-d(h)}$. Since $\#\mathcal{A}_n^+ = q^n$ for every $n \geq 0$, it follows that

$$\mathcal{B}(z, w') \ll_z \sum_{n \geq 0} \left[q|w'| \left(1 - \frac{q^{-2}}{|z|}\right)^{-1}\right]^n,$$

which converges for $|w'|$ sufficiently small.

Fix $P \in \mathcal{P}$ and write $Y = P^v Y_0$ with $v \geq 0$ and $P \nmid Y_0$, so that $Y^2 = P^{2v} Y_0^2$ and $\left(\frac{Y_0^2}{P}\right) = 1$. By (2.18) with $k = 2v$ and $V_1 = Y_0^2$, the values $\mathcal{G}(Y^2, \psi_{P^i})$ entering the P-Euler factor vanish for every $i \geq 1$ except

$$\mathcal{G}(Y^2, \psi_{P^{2j}}) = \phi(P^{2j}) \quad (1 \leq j \leq v), \qquad \mathcal{G}(Y^2, \psi_{P^{2v+1}}) = |P|^{2v} \sqrt{|P|}.$$

As in [6, Lemma 6.2], feeding them into the Euler-product computation — the finite sum over j summed first, then the geometric series over $v \geq 0$, and the result multiplied by the local factors $(1 - Z_P)$, $(1 - W_P)$ and $(1 - |P| W_P^2 Z_P)$ of $\mathcal{Z}_{\mathcal{A}}(z)^{-1}$, $\mathcal{Z}_{\mathcal{A}}(w')^{-1}$ and $\mathcal{Z}_{\mathcal{A}}(q(w')^2 z)^{-1}$, respectively — gives (4.6) with the same local factor (4.7), and $\prod_P \mathcal{B}_P(z, w')$ converges absolutely in (4.8).

Part (2) corresponds to [6, Lemma 6.3]. The local factors at P of $\mathcal{Z}_{\mathcal{A}}(w'/(q^2z))$ and $\mathcal{Z}_{\mathcal{A}}((w')^2)^{-1}$ are $\big(1-W_P/(|P|^2Z_P)\big)^{-1}$ and $1-W_P^2$, respectively, so (4.9) gives

$$\prod_{P\in\mathcal{P}}\mathcal{B}_P(z,w')=\mathcal{Z}_{\mathcal{A}}\!\left(\frac{w'}{q^2z}\right)\mathcal{Z}_{\mathcal{A}}((w')^2)^{-1}\prod_{P\in\mathcal{P}}\mathcal{D}_P(z,w'),$$

which turns (4.6) into (4.10) in the initial domain, and $\prod_P\mathcal{D}_P(z,w')$ converges absolutely and uniformly on compact subsets of (4.11). Since the remaining factors on the right-hand side of (4.10) are meromorphic there, that right-hand side provides the asserted meromorphic continuation of $\mathcal{B}(z,w')$. □

Finally we record the value of the Euler product $\prod_P\mathcal{B}_P(z,w')$ at $w'=q^{-1}$, in the variable u with $z=u^{-2}$, where it is given in terms of the Euler product $\mathcal{C}^*$ of (3.1).

Put $\tilde{u}:=u^2/q$, so that $u^{-2}=1/(q\tilde{u})$. By (4.7) the factors $\mathcal{B}_P$ are those of [6, Lemma 6.2], so the evaluation of $\prod_P\mathcal{B}_P\big(1/(q\tilde{u}),q^{-1}\big)$ in [6, Section 6], and the absolute convergence shown there for $q^{-1}<|\tilde{u}|<q$, apply here; in the variable u that range is $1<|u|<q$, and the evaluation reads, with $\mathcal{C}$ as in (3.2),

$$\prod_{P\in\mathcal{P}}\mathcal{B}_P(u^{-2},q^{-1})=\frac{\mathcal{C}(\tilde{u})}{\zeta_{\mathcal{A}}(2)}\,\frac{1-(q\tilde{u})^{-1}}{1-\tilde{u}}.$$

The only change here is the rescaling (3.2), which turns $\mathcal{C}(\tilde{u})=\mathcal{C}(u^2/q)$ into $\mathcal{C}^*(u^2)$; together with $1-(q\tilde{u})^{-1}=1-u^{-2}$ and $1-\tilde{u}=1-u^2/q$ it gives

$$\prod_{P\in\mathcal{P}}\mathcal{B}_P(u^{-2},q^{-1})=\frac{\mathcal{C}^*(u^2)}{\zeta_{\mathcal{A}}(2)}\,\frac{q(u^2-1)}{u^2(q-u^2)}\qquad(1<|u|<q).\tag{4.12}$$

Both sides of (4.12) are holomorphic on $1<|u|<q$: by (3.5) with $v=u^2$ its right-hand side equals $\zeta_{\mathcal{A}}(2)^{-1}\mathcal{K}(u^2)(u^2-1)u^{-2}$, and $\mathcal{K}(u^2)$ is holomorphic for $|u|<q$.

4.2. **Proof of Proposition 4.1.** Fix $N\in\{g,g-1\}$. Restricting the two cumulative sums of (2.30) to the nonzero squares $V=Y^2$ and using $\zeta_{\mathcal{A}}(2)^{-1}=(q-1)/q$ gives

$$s_{\mathrm{e}}(h;C)_{\square}=\frac{1}{\zeta_{\mathcal{A}}(2)}\left(q\sum_{Y\in\mathcal{A}^+_{\le\frac{d(h)}{2}-g+d(C)-2}}\frac{\mathcal{G}(Y^2,\psi_h)}{\sqrt{|h|}}-\sum_{Y\in\mathcal{A}^+_{\le\frac{d(h)}{2}-g+d(C)-1}}\frac{\mathcal{G}(Y^2,\psi_h)}{\sqrt{|h|}}\right),\tag{4.13}$$

while the same restriction in (2.31) gives

$$s_{\mathrm{o}}(h;C)_{\square}=\sum_{Y\in\mathcal{A}^+_{\frac{d(h)-3}{2}-g+d(C)}}\frac{\mathcal{G}(Y^2,\psi_h)}{\sqrt{|h|}}-\frac{1}{q}\sum_{Y\in\mathcal{A}^+_{\frac{d(h)-1}{2}-g+d(C)}}\frac{\mathcal{G}(Y^2,\psi_h)}{\sqrt{|h|}}.\tag{4.14}$$

We convert the Y-sums in (4.13) and (4.14) into contour integrals over $|z|=q^{-1-\kappa}$, where $\varepsilon>0$ is fixed and $0<\kappa<\min\{1/2,\varepsilon\}$. The even sector uses the cumulative form (2.12) of the function-field Perron formula and the odd sector the exact form (2.11), applied to the series $A_h(z)$ of (4.3); in each sector the two shifted Y-sums combine into the single factor $qz-1$, and we obtain

$$s_{\mathrm{e}}(h;C)_{\square}=\frac{1}{\zeta_{\mathcal{A}}(2)}\,\frac{1}{2\pi i}\oint_{|z|=q^{-1-\kappa}}\frac{(qz-1)A_h(z)}{(1-z)\,z^{\frac{d(h)}{2}-g+d(C)}}\,dz,\tag{4.15}$$

$$s_{\mathrm{o}}(h;C)_{\square}=\frac{1}{q}\,\frac{1}{2\pi i}\oint_{|z|=q^{-1-\kappa}}\frac{(qz-1)A_h(z)}{z^{\frac{d(h)+1}{2}-g+d(C)}}\,dz.\tag{4.16}$$

From here on the two sectors run in parallel, and we carry them together. For $\bullet\in\{\mathrm{e},\mathrm{o}\}$, let $\nu_\bullet\in\{0,1\}$ denote the value taken on the sector $\bullet$ by the parity $\nu(h)$ of §2.3, so that $\nu_{\mathrm{e}}=0$ and $\nu_{\mathrm{o}}=1$, and put

$$c_{\mathrm{e}}:=\frac{q^{2g+2}}{\zeta_{\mathcal{A}}(2)},\qquad c_{\mathrm{o}}:=q^{2g+3/2},\qquad\Theta_\bullet(z):=\frac{c_\bullet z^g(qz-1)}{(1-z)^{1-\nu_\bullet}}.$$

Inserting (4.15) into (2.32) and (4.16) into (2.34), whose prefactors q^{2g+2} and $q^{2g+5/2}$ are absorbed into c_{e} and c_{o}, gives, for $\bullet\in\{\mathrm{e},\mathrm{o}\}$,

$$\mathcal{S}^*_{N,\bullet}(V=\square)=\frac{1}{2\pi i}\oint_{|z|=q^{-1-\kappa}}\Theta_\bullet(z)\sum_{\substack{h,m\in\mathcal{A}^+\\ d(h)+d(m)\le N\\ \nu(h)=\nu_\bullet}}\frac{A_h(z)}{|h|\,|m|^{3/2}z^{\frac{d(h)+\nu_\bullet}{2}}}\sum_{\substack{C\in\mathcal{A}^+_{\le g}\\ C\mid h^\infty}}\frac{1}{|C|^2z^{d(C)}}\,dz.\tag{4.17}$$

We extend the C-sum in (4.17) to its full Euler product: for each $h \in \mathcal{A}^+$,

$$\sum_{\substack{C\in\mathcal{A}^+\\ C|h^\infty}}\frac{1}{|C|^2 z^{d(C)}} = \prod_{P|h}\Big(1-\frac{1}{|P|^2 z^{d(P)}}\Big)^{-1} - \sum_{\substack{C|h^\infty\\ d(C)\ge g+1}}\frac{1}{|C|^2 z^{d(C)}}. \tag{4.18}$$

One input changes in bounding the tail: (4.4) supplies the pointwise bound $|A_h(z)| \le (1-q|z|)^{-1} = (1-q^{-\kappa})^{-1} \ll_\kappa 1$, uniform in h, in place of the average used in [6, Section 6]. With it, for sufficiently small $\delta > 0$ the divisor bound for polynomials supported on the prime divisors of h gives, on $|z| = q^{-1-\kappa}$,

$$\sum_{\substack{C|h^\infty\\ d(C)\ge g+1}}\frac{1}{|C|^2|z|^{d(C)}} = \sum_{\substack{C|h^\infty\\ d(C)\ge g+1}} q^{-(1-\kappa)d(C)} \ll_{\kappa,\delta} q^{-(1-\kappa)g+\delta(g+d(h))},$$

and, using $d(h)+d(m) \le N \le g$ and $\sum_{m\in\mathcal{A}^+}|m|^{-3/2} \ll_q 1$, the tail contributes $\ll_{\kappa,\delta} q^{((1+\kappa)/2+2\delta)g}$ in either parity sector. Choosing $\delta > 0$ with $\kappa + 4\delta < \varepsilon$ makes this $O_\varepsilon\big(q^{g(1+\varepsilon)/2}\big)$. After the extension (4.18) the integrands are holomorphic in the annulus $q^{-3/2} \le |z| \le q^{-1-\kappa}$, since $A_h(z)$ is holomorphic for $|z| < q^{-1}$ while the possible poles of $\prod_{P|h}(1-|P|^{-2}z^{-d(P)})^{-1}$ lie on $|z| = q^{-2}$. The z-contour may therefore be moved to $|z| = q^{-3/2}$ without crossing a pole, and we obtain

$$\mathcal{S}^*_{N,\bullet}(V=\square) = \frac{1}{2\pi i}\oint_{|z|=q^{-3/2}}\Theta_\bullet(z)\widetilde{H}^\bullet_N(z)\,dz + O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big), \tag{4.19}$$

where

$$\widetilde{H}^\bullet_N(z) := \sum_{\substack{h,m\in\mathcal{A}^+\\ d(h)+d(m)\le N\\ \nu(h)=\nu_\bullet}}\frac{A_h(z)}{|h|\,|m|^{3/2}z^{\frac{d(h)+\nu_\bullet}{2}}}\prod_{P|h}\Big(1-\frac{1}{|P|^2z^{d(P)}}\Big)^{-1}.$$

In $\widetilde{H}^\bullet_N$ the constraint $d(h)+d(m) \le N$ is on the total degree, so w tracks $d(h)+d(m)$ and cannot select the parity of $d(h)$ alone; that parity has to be extracted from $\mathcal{B}$ instead. Accordingly, put

$$\mathcal{B}^\bullet(z,w') := \tfrac{1}{2}\big(\mathcal{B}(z,w') + (-1)^{\nu_\bullet}\mathcal{B}(z,-w')\big),$$

which selects the terms of (4.5) with $\nu(h) = \nu_\bullet$.

For the rest of this section, fix either square root $z^{1/2}$ of z and put $z^{3/2} := zz^{1/2}$; all half-integral powers of z refer to this choice. Set $w' = w/(qz^{1/2})$. Then $|h| = q^{d(h)}$ and $\#\mathcal{A}^+_n = q^n$ give

$$(w')^{d(h)} = \frac{w^{d(h)}}{|h|z^{\frac{d(h)}{2}}}, \qquad \sum_{m\in\mathcal{A}^+}\frac{w^{d(m)}}{|m|^{3/2}} = \frac{1}{1-q^{-1/2}w},$$

respectively, the second being the free divisor's contribution.

Define

$$K_{N,\bullet}(z,w) := \frac{z^{-\nu_\bullet/2}\,\mathcal{B}^\bullet\Big(z,\dfrac{w}{qz^{1/2}}\Big)}{(1-q^{-1/2}w)(1-w)w^{N+1}}.$$

The cumulative Perron formula (2.12) then gives

$$\widetilde{H}^\bullet_N(z) = \frac{1}{2\pi i}\oint_{|w|=r}K_{N,\bullet}(z,w)\,dw, \tag{4.20}$$

where $r > 0$ is sufficiently small; in particular, we may assume that $r < q^{-3/4}$, which under $w' = w/(qz^{1/2})$ and $|z| = q^{-3/2}$ reads $|w'| < q^{-1}$.

Lemma 4.4 (Poles of $K_{N,\bullet}$). *For $|z| = q^{-3/2}$, $K_{N,\bullet}(z,\cdot)$ is meromorphic on the disc $|w| < 1$, with no poles there other than the five points $w = 0$, $w = \pm z^{1/2}$ of modulus $q^{-3/4}$, and $w = \pm q^2 z^{3/2}$ of modulus $q^{-1/4}$.*

Proof. On $|z| = q^{-3/2}$ the substitution $w' = w/(qz^{1/2})$ gives $|w'| = q^{-1/4}|w|$, and for $|w| < 1$ the pair (z,w') satisfies the five conditions of (4.11), the binding one being $|w'|^2 < q|z|$, which reads $|w| < 1$; these conditions involve w' only through $|w'|$, so they hold for $-w'$ as well, hence for both terms of $\mathcal{B}^\bullet(z,w')$. By Lemma 4.3(2) the right-hand side of (4.10) therefore represents $\mathcal{B}(z,\pm w')$ on $|w| < 1$ as a meromorphic function whose only poles are those of the explicit zeta factors. As the remaining factors of $K_{N,\bullet}$ are rational in w, it follows that $K_{N,\bullet}(z,\cdot)$ is meromorphic on $|w| < 1$.

It remains to locate its poles. Inserting (4.10) into $\mathcal{B}^\bullet$ and undoing the substitution $w = qz^{1/2}w'$ displays every w-dependent factor of $K_{N,\bullet}$ at once: with $\varrho \in \{\pm 1\}$ indexing the two terms of $\mathcal{B}^\bullet$,

$$K_{N,\bullet}(z,w) = \frac{z^{-\nu_\bullet/2}\,\mathcal{Z}_{\mathcal{A}}(z)\left(1-\frac{w^2}{qz}\right)}{2\,(1-w^2)(1-w)(1-q^{-1/2}w)\,w^{N+1}} \sum_{\varrho\in\{\pm1\}} \frac{\varrho^{\nu_\bullet}\prod_{P\in\mathcal{P}}\mathcal{D}_P(z,\varrho w')}{\left(1-\frac{\varrho w}{z^{1/2}}\right)\left(1-\frac{\varrho w}{q^2z^{3/2}}\right)}.$$

Here the four zeta factors of (4.10) that involve w have been evaluated as follows, the last column recording the moduli on $|z| = q^{-3/2}$:

factor of (4.10)	in the variable w	singular at	$\lvert w\rvert$
$\mathcal{Z}_{\mathcal{A}}(\varrho w')$	$\left(1-\frac{\varrho w}{z^{1/2}}\right)^{-1}$	$w = \varrho z^{1/2}$	$q^{-3/4}$
$\mathcal{Z}_{\mathcal{A}}\left(\frac{\varrho w'}{q^2z}\right)$	$\left(1-\frac{\varrho w}{q^2z^{3/2}}\right)^{-1}$	$w = \varrho q^2 z^{3/2}$	$q^{-1/4}$
$\mathcal{Z}_{\mathcal{A}}\big(q(w')^2z\big)$	$\big(1-w^2\big)^{-1}$	$w = \pm 1$	1
$\mathcal{Z}_{\mathcal{A}}\big((w')^2\big)^{-1}$	$1-\frac{w^2}{qz}$	nowhere	

The Euler product $\prod_P \mathcal{D}_P(z,\varrho w')$ is holomorphic on $|w| < 1$ by Lemma 4.3(2), and the remaining factors $(1-w)^{-1}$, $(1-q^{-1/2}w)^{-1}$ and $w^{-(N+1)}$ of $K_{N,\bullet}$ are singular only at $w = 1$, $w = q^{1/2}$ and $w = 0$, respectively. Of all these singularities those at $w = \pm 1$ and $w = q^{1/2}$ have modulus ≥ 1, so the only poles in $|w| < 1$ are $w = 0$, $w = \pm z^{1/2}$ and $w = \pm q^2 z^{3/2}$. □

4.2.1. *Shifting the w-contour.* We now enlarge the circle $|w| = r$ in (4.20) to $|w| = q^{-\beta}$: fix $\varepsilon > 0$ and $0 < \beta < \min\{1/4, \varepsilon/2\}$, so that $r < q^{-3/4} < q^{-1/4} < q^{-\beta} < 1$. By Lemma 4.4 the enlargement therefore crosses exactly the four nonzero poles $w = \pm z^{1/2}$ and $w = \pm q^2 z^{3/2}$, while $w = 0$ is enclosed by both contours.

Nothing depends on the choice of $z^{1/2}$: replacing it by $-z^{1/2}$ replaces w' by $-w'$, which leaves $\mathcal{B}^{\mathrm{e}}$ unchanged and changes the sign of both $z^{-1/2}$ and $\mathcal{B}^{\mathrm{o}}$, so $K_{N,\bullet}$ is unchanged; and the two crossed pairs $\{\pm z^{1/2}\}$ and $\{\pm q^2 z^{3/2}\}$ are unchanged as sets, so the residue sums over them are unchanged as well.

By Lemma 4.4 and the residue theorem, for $\bullet \in \{\mathrm{e}, \mathrm{o}\}$ and $|z| = q^{-3/2}$,

$$\begin{aligned}\frac{1}{2\pi i}\oint_{|w|=r} K_{N,\bullet}(z,w)\,dw = \frac{1}{2\pi i}\oint_{|w|=q^{-\beta}} K_{N,\bullet}(z,w)\,dw \\ - \sum_{\omega\in\{\pm z^{1/2}\}} \mathrm{Res}_{w=\omega}\,K_{N,\bullet}(z,w) - \sum_{\omega\in\{\pm q^2 z^{3/2}\}} \mathrm{Res}_{w=\omega}\,K_{N,\bullet}(z,w).\end{aligned} \tag{4.21}$$

Substituting (4.21) into (4.20) and the result into (4.19), we obtain

$$\mathcal{S}^*_{N,\bullet}(V=\square) = A^*_{N,\bullet} + B^*_{N,\bullet} + C^*_{N,\bullet} + O_\varepsilon\left(q^{g(1+\varepsilon)/2}\right),$$

where

$$A^*_{N,\bullet} := -\frac{1}{2\pi i}\oint_{|z|=q^{-3/2}} \Theta_\bullet(z) \sum_{\omega\in\{\pm z^{1/2}\}} \mathrm{Res}_{w=\omega}\,K_{N,\bullet}(z,w)\,dz, \tag{4.22}$$

$$B^*_{N,\bullet} := -\frac{1}{2\pi i}\oint_{|z|=q^{-3/2}} \Theta_\bullet(z) \sum_{\omega\in\{\pm q^2 z^{3/2}\}} \mathrm{Res}_{w=\omega}\,K_{N,\bullet}(z,w)\,dz, \tag{4.23}$$

$$C^*_{N,\bullet} := \frac{1}{(2\pi i)^2}\oint_{|z|=q^{-3/2}}\oint_{|w|=q^{-\beta}} \Theta_\bullet(z) K_{N,\bullet}(z,w)\,dw\,dz. \tag{4.24}$$

Thus $A^*_{N,\bullet}$ and $B^*_{N,\bullet}$ collect the residues at $w = \pm z^{1/2}$ and at $w = \pm q^2 z^{3/2}$, respectively, while $C^*_{N,\bullet}$ is the integral over the enlarged contour; the minus signs in (4.22) and (4.23) come from those in (4.21).

The residue terms $A^*_{N,\bullet}$ and $B^*_{N,\bullet}$ are evaluated in §4.2.2 and §4.2.3, respectively. The enlarged-contour terms are error terms (cf. [10, (3.22)] and [3, (5.8) and (5.12)]); we record the verification because $K_{N,\bullet}$ here carries the additional factors $(1-q^{-1/2}w)^{-1}$ and $(1-w)^{-1}$. On $|z| = q^{-3/2}$ and $|w| = q^{-\beta}$ the four zeta factors of (4.10) that occur with positive power have arguments of moduli

$$|z| = q^{-3/2}, \qquad |\pm w'| = q^{-1/4-\beta}, \qquad |q(w')^2 z| = q^{-1-2\beta}, \qquad \left|\frac{\pm w'}{q^2 z}\right| = q^{-3/4-\beta},$$

none of which equals q^{-1} when $0 < \beta < 1/4$; with the locally uniform convergence of $\prod_P \mathcal{D}_P$ on (4.11) this gives $\mathcal{B}^{\mathrm{e}}, \mathcal{B}^{\mathrm{o}} \ll_\beta 1$ there. The remaining factors of $K_{N,\bullet}$ are bounded as well, $|1-w|$ and $|1-q^{-1/2}w|$ being bounded away from zero on $|w| = q^{-\beta}$ while $|w^{-(N+1)}| = q^{\beta(N+1)}$ and $|z^{-1/2}| = q^{3/4}$, so that $K_{N,\bullet}(z,w) \ll_\beta q^{\beta(N+1)}$.

On $|z| = q^{-3/2}$ one has $\Theta_\bullet(z) \ll_q q^{2g+2}|z|^g$, the factors $(qz-1)$ and $(1-z)^{\nu_\bullet - 1}$ being $\ll_q 1$ there; estimating the z-integral in (4.24) trivially, with $|z^g| = q^{-3g/2}$ and contour length $\ll q^{-3/2}$, we obtain

$$C^*_{N,\bullet} \ll_\beta q^{2g+2} \cdot q^{-3g/2} \cdot q^{-3/2} \cdot q^{\beta(N+1)} \ll_\beta q^{g/2+\beta N}.$$

Since $N \le g$ and $\beta < \varepsilon/2$, it follows that $C^*_{N,\bullet} = O_\varepsilon\big(q^{g(1+\varepsilon)/2}\big)$. Consequently,

$$\mathcal{S}^*_{N,\bullet}(V=\square) = A^*_{N,\bullet} + B^*_{N,\bullet} + O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big). \tag{4.25}$$

Summing (4.25) over the two sectors, we obtain

$$\mathcal{S}^*_{N,\mathrm{e}}(V=\square) + \mathcal{S}^*_{N,\mathrm{o}}(V=\square) = A^*_{N,\mathrm{e}} + A^*_{N,\mathrm{o}} + B^*_{N,\mathrm{e}} + B^*_{N,\mathrm{o}} + O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big). \tag{4.26}$$

4.2.2. *Principal square-dual residues and parity matching.* This subsection evaluates $A^*_{N,\bullet}$, the contribution of the pole family $w = \pm z^{1/2}$. The outcome, Proposition 4.6, is that its total over $N \in \{g, g-1\}$ and over the two parity sectors equals $-\mathcal{R}_0(g;\rho)$, and so cancels the contour integral retained in §3.

Lemma 4.5 (Residues at $w = \pm z^{1/2}$)**.** *For $|z| = q^{-3/2}$, one has*

$$\sum_{\omega \in \{\pm z^{1/2}\}} \mathrm{Res}_{w=\omega} K_{N,\bullet}(z,w) = -\frac{\mathcal{Z}_{\mathcal{A}}(z)\mathcal{Z}_{\mathcal{A}}(z/q)}{2z^{(N+\nu_\bullet)/2}} R^\bullet_N(z) \prod_{P\in\mathcal{P}} \mathcal{B}_P(z, q^{-1}), \tag{4.27}$$

where

$$R^\bullet_N(z) := \frac{1}{(1-q^{-1/2}z^{1/2})(1-z^{1/2})} + \frac{(-1)^{N+\nu_\bullet}}{(1+q^{-1/2}z^{1/2})(1+z^{1/2})}.$$

Proof. Write $w' = w/(qz^{1/2})$, so that $w = \pm z^{1/2}$ corresponds to $w' = \pm q^{-1}$. For $|z| = q^{-3/2}$ and $|w'| = q^{-1}$, the three conditions in (4.8) hold, so (4.6) is available near $w = \pm z^{1/2}$ and $\prod_P \mathcal{B}_P(z,w')$ is holomorphic there: the pole is carried by $\mathcal{Z}_{\mathcal{A}}(w')$ alone. At $w = z^{1/2}$ this factor is $\mathcal{Z}_{\mathcal{A}}(w/(qz^{1/2})) = (1 - w/z^{1/2})^{-1}$, a simple pole of residue $-z^{1/2}$, while the other three factors of (4.6), namely $\mathcal{Z}_{\mathcal{A}}(z)$, $\mathcal{Z}_{\mathcal{A}}(q(w')^2 z)$ and $\prod_P \mathcal{B}_P(z,w')$, are evaluated at $w' = q^{-1}$, where $q(w')^2 z = z/q$. Hence

$$\mathrm{Res}_{w=z^{1/2}} \mathcal{B}\left(z, \frac{w}{qz^{1/2}}\right) = -z^{1/2}\,\Pi(z), \qquad \text{where} \quad \Pi(z) := \mathcal{Z}_{\mathcal{A}}(z)\mathcal{Z}_{\mathcal{A}}(z/q) \prod_{P\in\mathcal{P}} \mathcal{B}_P(z, q^{-1}).$$

At $w = -z^{1/2}$ the same computation applies to $\mathcal{B}(z, -w/(qz^{1/2}))$: the singular factor is $\mathcal{Z}_{\mathcal{A}}(-w/(qz^{1/2})) = (1 + w/z^{1/2})^{-1}$, of residue $+z^{1/2}$, whence $\mathrm{Res}_{w=-z^{1/2}} \mathcal{B}(z, -w/(qz^{1/2})) = +z^{1/2}\Pi(z)$.

At each of the two points $w = \pm z^{1/2}$ only one of the two terms of $\mathcal{B}^\bullet$ is singular, so

$$\mathrm{Res}_{w=z^{1/2}} \mathcal{B}^\bullet\left(z, \frac{w}{qz^{1/2}}\right) = -\tfrac{1}{2} z^{1/2}\Pi(z), \qquad \mathrm{Res}_{w=-z^{1/2}} \mathcal{B}^\bullet\left(z, \frac{w}{qz^{1/2}}\right) = (-1)^{\nu_\bullet}\tfrac{1}{2} z^{1/2}\Pi(z),$$

the factor $\frac{1}{2}$ and the sign $(-1)^{\nu_\bullet}$ coming from the definition of $\mathcal{B}^\bullet$. Since $(1-q^{-1/2}w)(1-w)w^{N+1}$ is holomorphic and nonzero at $w = \pm z^{1/2}$, the definition of $K_{N,\bullet}$ gives

$$\mathrm{Res}_{w=\omega} K_{N,\bullet}(z,w) = \frac{z^{-\nu_\bullet/2}}{(1-q^{-1/2}\omega)(1-\omega)\,\omega^{N+1}} \mathrm{Res}_{w=\omega} \mathcal{B}^\bullet\left(z, \frac{w}{qz^{1/2}}\right) \qquad (\omega \in \{\pm z^{1/2}\}).$$

Summing the two, and using $\big(-z^{1/2}\big)^{-(N+1)} = (-1)^{N+1} z^{-(N+1)/2}$, we obtain

$$\sum_{\omega\in\{\pm z^{1/2}\}} \mathrm{Res}_{w=\omega} K_{N,\bullet}(z,w) = -\frac{z^{-\nu_\bullet/2}\Pi(z)}{2z^{N/2}} \left\{ \frac{1}{(1-q^{-1/2}z^{1/2})(1-z^{1/2})} + \frac{(-1)^{N+\nu_\bullet}}{(1+q^{-1/2}z^{1/2})(1+z^{1/2})} \right\}.$$

Here the sign $(-1)^{N+\nu_\bullet}$ originates in $w^{-(N+1)}$ at the negative pole. The sum in braces is $R^\bullet_N(z)$, and $z^{\nu_\bullet/2} z^{N/2} = z^{(N+\nu_\bullet)/2}$; this proves (4.27). □

We shall pass from z to the variable u with $z = u^{-2}$, in which $\mathcal{R}_0(g;\rho)$ is defined in (3.8). If $R > 0$ and G is holomorphic in a neighborhood of $|z| = R^2$, then substituting $z = s^2$, the circle $|s| = R$ covering $|z| = R^2$ twice, and then $s = u^{-1}$, which reverses the orientation and so cancels the sign in $ds = -u^{-2}\,du$, gives

$$\frac{1}{2\pi i}\oint_{|z|=R^2} G(z)\,dz = \frac{1}{2\pi i}\oint_{|s|=R} G(s^2)\,s\,ds = \frac{1}{2\pi i}\oint_{|u|=R^{-1}} u^{-3} G(u^{-2})\,du. \tag{4.28}$$

Substituting the residue formula of Lemma 4.5 into (4.22), so that the minus signs of (4.22) and (4.27) cancel, and using $(qz-1)\mathcal{Z}_{\mathcal{A}}(z) = -1$ and $\mathcal{Z}_{\mathcal{A}}(z/q) = (1-z)^{-1}$, which together give $\Theta_\bullet(z)\mathcal{Z}_{\mathcal{A}}(z)\mathcal{Z}_{\mathcal{A}}(z/q) = -c_\bullet z^g (1-z)^{\nu_\bullet - 2}$, we obtain

$$A^*_{N,\bullet} = -\frac{c_\bullet}{2}\frac{1}{2\pi i}\oint_{|z|=q^{-3/2}} \frac{z^{g-\frac{N+\nu_\bullet}{2}}}{(1-z)^{2-\nu_\bullet}} R^\bullet_N(z) \prod_{P\in\mathcal{P}} \mathcal{B}_P(z, q^{-1})\,dz.$$

We now convert this into an integral in the variable u: we apply (4.28) with $R = q^{-3/4}$, taking $z = u^{-2}$ and $z^{1/2} = u^{-1}$; the new contour is $|u| = q^{3/4}$, which lies in the annulus $1 < |u| < q$ where (4.12) is available. Under this substitution

$$R_N^\bullet(u^{-2}) = u^2 \left(\frac{1}{(u - q^{-1/2})(u-1)} + \frac{(-1)^{N+\nu_\bullet}}{(u+q^{-1/2})(u+1)} \right).$$

Using (4.12) we obtain

$$A_{N,\bullet}^* = -\frac{q\, c_\bullet}{2\zeta_{\mathcal{A}}(2)} \frac{1}{2\pi i} \oint_{|u|=q^{3/4}} \frac{u^{N-2g+1-\nu_\bullet}\, \mathcal{C}^*(u^2)}{(u^2-1)^{1-\nu_\bullet}(q-u^2)} \left(\frac{1}{(u - q^{-1/2})(u-1)} + \frac{(-1)^{N+\nu_\bullet}}{(u+q^{-1/2})(u+1)} \right) du. \tag{4.29}$$

Proposition 4.6. *For every $1 < \rho < q$, one has*

$$\sum_{N \in \{g, g-1\}} \left(A_{N,\mathrm{e}}^* + A_{N,\mathrm{o}}^* \right) = -\mathcal{R}_0(g;\rho), \tag{4.30}$$

with $\mathcal{R}_0(g;\rho)$ as in (3.8).

Proof. By (3.5) with $v = u^2$, the integrand in (4.29) is holomorphic in the annulus $1 < |u| < q$; its contour may therefore be moved to $|u| = \rho$ for any $1 < \rho < q$. We do so for each of the four pairs $(N, \bullet)$ with $N \in \{g, g-1\}$ and $\bullet \in \{\mathrm{e}, \mathrm{o}\}$, and add the two integrals in the definition (3.8) of $\mathcal{R}_0(g;\rho)$. Since $(-1)^{N+\nu_\bullet}$ equals $(-1)^g$ for (g, e) and $(g-1, \mathrm{o})$ and $-(-1)^g$ for the other two, and using $\zeta_{\mathcal{A}}(2)^{-1} = (q-1)/q$, the resulting six integrands — four from (4.29) and two from (3.8) — may be placed over a common denominator. Factoring out $q^{2g+2}\mathcal{C}^*(u^2)u^{-g-1}$ and treating the two parities of g separately, a direct computation gives

$$\sum_{N \in \{g, g-1\}} \left(A_{N,\mathrm{e}}^* + A_{N,\mathrm{o}}^* \right) + \mathcal{R}_0(g;\rho) = \frac{q^{2g+2}}{2\pi i} \oint_{|u|=\rho} \frac{\mathcal{C}^*(u^2)}{u^{g+1}} \mathcal{E}_g(u)\, du, \tag{4.31}$$

where

$$\mathcal{E}_g(u) := \begin{cases} \dfrac{(q-1)u(u^2+1+2\sqrt{q})}{\sqrt{q}\,(q-u^2)(u^2-1)^2}, & g \text{ even}, \\[2ex] \dfrac{(q-1)(\sqrt{q}(1+u^2)+2u^2)}{\sqrt{q}\,(q-u^2)(u^2-1)^2}, & g \text{ odd}. \end{cases}$$

The integrand on the right of (4.31) is holomorphic on $1 < |u| < q$, since $(u^2-1)^{-2}$ is singular only on $|u| = 1$ and $\mathcal{C}^*(u^2)/(q-u^2)$ is holomorphic there by (3.5); and it is an even function of u, because $\mathcal{C}^*(u^2)$ is even while u^{-g-1} and $\mathcal{E}_g(u)$ are both odd when g is even and both even when g is odd. The Laurent coefficients of an even function on an annulus satisfy $c_n = (-1)^n c_n$, so $c_{-1} = 0$ and the integral vanishes; thus (4.31) yields (4.30). □

4.2.3. *Secondary square-dual poles.* This subsection evaluates $B_{N,\bullet}^*$, the contribution of the pole family $w = \pm q^2 z^{3/2}$. The outcome, Proposition 4.8, is that its total over $N \in \{g, g-1\}$ and over the two parity sectors equals, up to an admissible error, the sum $\mathcal{S}_0(g)$ of the residues at the three cube roots of q^{-4} crossed when the z-contour is pushed out; all three are retained here, whereas the earlier treatments keep only the real one (see Remark 6.1).

Lemma 4.7 (Residues at $w = \pm q^2 z^{3/2}$)**.** *For $|z| = q^{-3/2}$, one has*

$$\sum_{\omega \in \{\pm q^2 z^{3/2}\}} \mathrm{Res}_{w=\omega} K_{N,\bullet}(z,w) = -\frac{\mathcal{Z}_{\mathcal{A}}(z)\mathcal{Z}_{\mathcal{A}}(qz)\mathcal{Z}_{\mathcal{A}}(q^3z^3)}{2q^{2N} z^{(3N+\nu_\bullet)/2} \mathcal{Z}_{\mathcal{A}}(q^2z^2)} Q_N^\bullet(z) \prod_{P \in \mathcal{P}} \mathcal{D}_P(z, qz), \tag{4.32}$$

where

$$Q_N^\bullet(z) := \frac{1}{(1-q^{3/2}z^{3/2})(1-q^2z^{3/2})} + \frac{(-1)^{N+\nu_\bullet}}{(1+q^{3/2}z^{3/2})(1+q^2z^{3/2})}. \tag{4.33}$$

Proof. Set $a := q^2 z^{3/2}$, the value of w at which $w' = w/(qz^{1/2})$ equals qz. In (4.10) the factor $\mathcal{Z}_{\mathcal{A}}(w'/(q^2z))$ becomes $(1 - w/a)^{-1}$, a simple pole at $w = a$ of residue $-a$. On $|z| = q^{-3/2}$ the pair (z, qz) satisfies the five conditions of (4.11). At $w = a$, that is $w' = qz$, the three remaining zeta factors $\mathcal{Z}_{\mathcal{A}}(z)$, $\mathcal{Z}_{\mathcal{A}}(w')$ and $\mathcal{Z}_{\mathcal{A}}(q(w')^2 z)$ of (4.10) have the arguments z, qz and q^3z^3, none of which equals q^{-1}; they are therefore holomorphic there, and their value, together with $\mathcal{Z}_{\mathcal{A}}((w')^2)^{-1} = \mathcal{Z}_{\mathcal{A}}(q^2z^2)^{-1}$ and $\prod_P \mathcal{D}_P(z, qz)$, is

$$\mathscr{F}(z) := \frac{\mathcal{Z}_{\mathcal{A}}(z)\mathcal{Z}_{\mathcal{A}}(qz)\mathcal{Z}_{\mathcal{A}}(q^3z^3)}{\mathcal{Z}_{\mathcal{A}}(q^2z^2)} \prod_{P \in \mathcal{P}} \mathcal{D}_P(z, qz).$$

Hence $\mathrm{Res}_{w=a} \mathcal{B}(z, w/(qz^{1/2})) = -a\mathscr{F}(z)$, and replacing w by $-w$, which reverses the sign of the residue, gives $\mathrm{Res}_{w=-a} \mathcal{B}(z, -w/(qz^{1/2})) = a\mathscr{F}(z)$.

The rest is the computation in the proof of Lemma 4.5, with $z^{1/2}$ replaced by a and $\Pi(z)$ by $\mathscr{F}(z)$: at each of $w = \pm a$ only one of the two terms of $\mathcal{B}^\bullet$ is singular, so its residue there is one half of the corresponding value

above, that at $w=-a$ carrying the extra factor $(-1)^{\nu_\bullet}$ from the definition of $\mathcal{B}^\bullet$; and since $(1-q^{-1/2}w)(1-w)w^{N+1}$ is holomorphic and nonzero at $w=\pm a$,

$$\operatorname{Res}_{w=\omega} K_{N,\bullet}(z,w)=\frac{z^{-\nu_\bullet/2}}{(1-q^{-1/2}\omega)(1-\omega)\,\omega^{N+1}}\operatorname{Res}_{w=\omega}\mathcal{B}^\bullet\left(z,\frac{w}{qz^{1/2}}\right)\qquad(\omega\in\{\pm a\}).$$

Summing the two, and using $(-a)^{-(N+1)}=(-1)^{N+1}a^{-(N+1)}$, we obtain

$$\sum_{\omega\in\{\pm a\}}\operatorname{Res}_{w=\omega}K_{N,\bullet}(z,w)=-\frac{z^{-\nu_\bullet/2}\mathscr{F}(z)}{2a^N}\left\{\frac{1}{(1-q^{-1/2}a)(1-a)}+\frac{(-1)^{N+\nu_\bullet}}{(1+q^{-1/2}a)(1+a)}\right\}.$$

Since $q^{-1/2}a=q^{3/2}z^{3/2}$ the sum in braces is $Q_N^\bullet(z)$, and $z^{\nu_\bullet/2}a^N=q^{2N}z^{(3N+\nu_\bullet)/2}$; this proves (4.32). □

The factor $(qz-1)$ carried by $\Theta_\bullet$ in (4.23) combines with the four zeta factors of (4.32):

$$(qz-1)\mathcal{Z}_{\mathcal{A}}(z)\frac{\mathcal{Z}_{\mathcal{A}}(qz)\mathcal{Z}_{\mathcal{A}}(q^3z^3)}{\mathcal{Z}_{\mathcal{A}}(q^2z^2)}=-\frac{1-q^3z^2}{(1-q^2z)(1-q^4z^3)}.\tag{4.34}$$

Substituting (4.32) into (4.23), so that the minus signs of (4.23) and (4.32) cancel, and using (4.34), we obtain

$$B^*_{N,\bullet}=-\frac{c_\bullet}{2q^{2N}}\frac{1}{2\pi i}\oint_{|z|=q^{-3/2}}J_{N,\bullet}(z)\,dz,\tag{4.35}$$

where

$$J_{N,\bullet}(z):=z^{g-\frac{3N+\nu_\bullet}{2}}\frac{(1-q^3z^2)\prod_{P\in\mathcal{P}}\mathcal{D}_P(z,qz)}{(1-z)^{1-\nu_\bullet}(1-q^2z)(1-q^4z^3)}Q_N^\bullet(z).\tag{4.36}$$

Write $A:=q^{3/2}z^{3/2}$ and $B:=q^2z^{3/2}$, so that $A^2=q^3z^3$, $B^2=q^4z^3$ and $AB=q^{7/2}z^3$. Putting the two fractions of (4.33) over the common denominator $(1-A^2)(1-B^2)$ gives

$$\frac{1}{(1-A)(1-B)}+\frac{1}{(1+A)(1+B)}=\frac{2(1+q^{7/2}z^3)}{(1-q^3z^3)(1-q^4z^3)},\tag{4.37}$$

$$\frac{1}{(1-A)(1-B)}-\frac{1}{(1+A)(1+B)}=\frac{2(q^{3/2}+q^2)z^{3/2}}{(1-q^3z^3)(1-q^4z^3)}.\tag{4.38}$$

Thus $Q_N^\bullet(z)$ is given by (4.37) when $N+\nu_\bullet$ is even and by (4.38) when it is odd. In either case the half-integral powers cancel against $z^{g-(3N+\nu_\bullet)/2}$ in (4.36), because $3N+\nu_\bullet\equiv N+\nu_\bullet \bmod 2$; thus $J_{N,\bullet}(z)$ is a single-valued function of z.

For $q^{-2}<|z|<q^{-1}$, the substitution $w'=qz$ satisfies all the conditions in (4.11), so $\prod_{P\in\mathcal{P}}\mathcal{D}_P(z,qz)$ is holomorphic throughout this annulus. After substituting (4.37) and (4.38) into (4.36), the denominator of $J_{N,\bullet}$ is a product of factors among $1-z$, $1-q^2z$, $1-q^3z^3$ and $(1-q^4z^3)^2$, the last squared because $(1-q^4z^3)^{-1}$ occurs both in (4.36) and in (4.37)–(4.38). The first three vanish only at $z=1$, at $z=q^{-2}$ and on $|z|=q^{-1}$, none of which lies in the open annulus, whereas the zeros of $1-q^4z^3$ are $z_k:=q^{-4/3}e^{2\pi ik/3}$ $(k\in\{0,1,2\})$ with $q^{-2}<|z_k|=q^{-4/3}<q^{-1}$. Thus z_0, z_1, z_2 are the only poles of $J_{N,\bullet}$ in $q^{-2}<|z|<q^{-1}$, and each has order at most two, the zeros of $1-q^4z^3$ being simple.

Define the total residue contribution at z_0, z_1, z_2 by

$$\mathcal{S}_0(g):=\sum_{N\in\{g,g-1\}}\sum_{\bullet\in\{\mathrm{e},\mathrm{o}\}}\sum_{k=0}^{2}\frac{c_\bullet}{2q^{2N}}\operatorname{Res}_{z=z_k}J_{N,\bullet}(z).\tag{4.39}$$

Proposition 4.8. *For every $\varepsilon>0$, one has*

$$\sum_{N\in\{g,g-1\}}\left(B^*_{N,\mathrm{e}}+B^*_{N,\mathrm{o}}\right)=\mathcal{S}_0(g)+O_\varepsilon\left(q^{g(1+\varepsilon)/2}\right).\tag{4.40}$$

Proof. Fix $\varepsilon>0$ and $0<\delta<\min\{1/3,\varepsilon\}$, and move the contours in (4.35) from $|z|=q^{-3/2}$ to $|z|=q^{-1-\delta}$. Since $q^{-3/2}<|z_k|=q^{-4/3}<q^{-1-\delta}<q^{-1}$, the only singularities of $J_{N,\bullet}$ that this shift can cross are $z=z_k$ $(k\in\{0,1,2\})$; so by the residue theorem and (4.39),

$$\sum_{N\in\{g,g-1\}}\left(B^*_{N,\mathrm{e}}+B^*_{N,\mathrm{o}}\right)=\mathcal{S}_0(g)-\sum_{N\in\{g,g-1\}}\sum_{\bullet\in\{\mathrm{e},\mathrm{o}\}}\frac{c_\bullet}{2q^{2N}}\frac{1}{2\pi i}\oint_{|z|=q^{-1-\delta}}J_{N,\bullet}(z)\,dz.$$

Write $N=g-j$ with $j\in\{0,1\}$. On $|z|=q^{-1-\delta}$ the prefactor of (4.35), combined with the explicit power of z in (4.36), is

$$c_\bullet q^{-2N}|z|^{g-\frac{3N+\nu_\bullet}{2}}=q^{(1+\delta)g/2+2+\delta\nu_\bullet/2+(1-3\delta)j/2},$$

which for $0<\delta<1/3$, $j\in\{0,1\}$ and $\nu_\bullet\in\{0,1\}$ is $\ll_\delta q^{(1+\delta)g/2}$. The circle $|z|=q^{-1-\delta}$ is compactly contained in $q^{-2}<|z|<q^{-1}$, so $\prod_P\mathcal{D}_P(z,qz)\ll_\delta 1$ there by Lemma 4.3, the remaining rational factors are holomorphic in a neighbourhood of the circle by (4.37) and (4.38), and the length of the circle does not depend on g. The outer integrals therefore contribute $O_\delta(q^{(1+\delta)g/2})$, and $\delta<\varepsilon$ gives (4.40). □

4.2.4. *Explicit evaluation of the secondary term.* This subsection evaluates the residue sum $\mathcal{S}_0(g)$ defined in (4.39) and proves the explicit periodic formula for it promised in Proposition 4.1. Throughout we work in the square-root coordinate $z = s^2$, which replaces the half-integral powers $z^{3/2}$ in $Q_N^\bullet(z)$ by the ordinary powers s^3.

For each $k \in \{0,1,2\}$, choose the canonical lift $s_k := q^{-2/3}e^{-2\pi ik/3}$ of z_k, so that $s_k^2 = z_k$ and $s_k^3 = q^{-2}$. The two lifts of z_k are then ϱs_k with $\varrho \in \{\pm 1\}$, and $(\varrho s_k)^2 = z_k$, $(\varrho s_k)^3 = \varrho q^{-2}$. This choice is used throughout the subsection.

In the coordinate s the two terms of (4.33) are indexed uniformly by a sign $\eta \in \{\pm 1\}$:

$$Q_N^\bullet(s^2) = \sum_{\eta\in\{\pm1\}} \frac{\eta^{N+\nu_\bullet}}{(1-\eta q^{3/2}s^3)(1-\eta q^2 s^3)}. \tag{4.41}$$

Since $z_k \neq 0$, the map $s \mapsto s^2$ is locally biholomorphic at each of the two lifts $\pm s_k$ of z_k, so for F meromorphic near z_k the change-of-variables formula for residues gives $\operatorname{Res}_{s=\varrho s_k}\big(2sF(s^2)\big) = \operatorname{Res}_{z=z_k} F(z)$ for either ϱ. Averaging over both lifts,

$$\operatorname{Res}_{z=z_k} F(z) = \frac{1}{2}\sum_{\varrho\in\{\pm1\}} \operatorname{Res}_{s=\varrho s_k}\big(2sF(s^2)\big), \tag{4.42}$$

a symmetric form that puts $\pm s_0$, $\pm s_1$, $\pm s_2$ on an equal footing; it is what makes the residues at s_k and at $-s_k$ agree in Lemma 4.9.

We now express $J_{N,\bullet}$, which appears in (4.39), in the square-root coordinate. From (4.36) and (4.41),

$$2sJ_{N,\bullet}(s^2) = 2s^{2g-3N+1-\nu_\bullet}\frac{(1-q^3s^4)\prod_{P\in\mathcal{P}}\mathcal{D}_P(s^2,qs^2)}{(1-s^2)^{1-\nu_\bullet}(1-q^2s^2)(1-q^4s^6)}\sum_{\eta\in\{\pm1\}}\frac{\eta^{N+\nu_\bullet}}{(1-\eta q^{3/2}s^3)(1-\eta q^2s^3)}. \tag{4.43}$$

We next determine which η-summand of (4.43) can be singular at $s = \varrho s_k$. Fix $k \in \{0,1,2\}$ and $\varrho \in \{\pm 1\}$; the sign η continues to index the two summands in (4.43), whereas ϱ specifies one of the two lifts of z_k.

The term of (4.43) indexed by η has, in the even sector, denominator

$$(1-s^2)(1-q^2s^2)(1-q^4s^6)(1-\eta q^{3/2}s^3)(1-\eta q^2s^3), \tag{4.44}$$

and in the odd sector the same denominator without the factor $1-s^2$. Since $(\varrho s_k)^2 = z_k$ and $(\varrho s_k)^3 = \varrho q^{-2}$, among the four factors of (4.44) other than $1-q^4s^6$ only $1-\eta q^2s^3$ can vanish at $s = \varrho s_k$, and it does so exactly when $\eta = \varrho$. The remaining factor $1-q^4s^6$ is common to the two η-terms and has a simple zero at ϱs_k. Thus for $\eta = \varrho$ two of the five factors in (4.44) vanish and the term has a pole of order at most two, whereas for $\eta = -\varrho$ only $1-q^4s^6$ vanishes. That remaining simple pole is spurious: adding the contributions of $N = g$ and $N = g-1$ produces a factor $1+\eta q^2s^3$, and $(1+\eta q^2s^3)/(1-q^4s^6) = (1-\eta q^2s^3)^{-1}$, which equals $\frac{1}{2}$ at $s = \varrho s_k$ when $\eta = -\varrho$. This is carried out in Lemma 4.9; accordingly, for $\eta \in \{\pm 1\}$ put

$$T_\eta(s) := \eta^g s^{-g}\frac{q(1-q^3s^4)\prod_{P\in\mathcal{P}}\mathcal{D}_P(s^2,qs^2)}{2(1-q^2s^2)(1-\eta q^{3/2}s^3)(1-\eta q^2s^3)^2}\left(\frac{(q-1)s}{1-s^2}+\sqrt{q}\,\eta\right). \tag{4.45}$$

Since $\prod_P \mathcal{D}_P(s^2,qs^2)$ depends only on s^2, replacing s by $-s$ and η by $-\eta$ in (4.45) gives

$$T_{-\eta}(-s) = -T_\eta(s). \tag{4.46}$$

Lemma 4.9. *One has*

$$\mathcal{S}_0(g) = 2\sum_{k=0}^{2}\operatorname{Res}_{s=s_k} T_{+1}(s). \tag{4.47}$$

Proof. Applying (4.42) to the residues in (4.39) and substituting (4.43), the factor $\frac{1}{2}$ of (4.42) cancelling the leading 2 there, we obtain by linearity of the residue

$$\mathcal{S}_0(g) = \sum_{k=0}^{2}\sum_{\varrho\in\{\pm1\}}\sum_{\eta\in\{\pm1\}}\operatorname{Res}_{s=\varrho s_k}\Bigg[\frac{(1-q^3s^4)\prod_{P\in\mathcal{P}}\mathcal{D}_P(s^2,qs^2)}{(1-q^2s^2)(1-q^4s^6)(1-\eta q^{3/2}s^3)(1-\eta q^2s^3)} \times \sum_{N\in\{g,g-1\}}\sum_{\bullet\in\{\mathrm{e},\mathrm{o}\}}\frac{c_\bullet}{2q^{2N}}\frac{\eta^{N+\nu_\bullet}s^{2g-3N+1-\nu_\bullet}}{(1-s^2)^{1-\nu_\bullet}}\Bigg]. \tag{4.48}$$

Fix η and evaluate the double sum over $N \in \{g, g-1\}$ and $\bullet \in \{\mathrm{e},\mathrm{o}\}$ in (4.48) term by term; using $c_\mathrm{e} = q^{2g+2}/\zeta_{\mathcal{A}}(2) = q^{2g+1}(q-1)$, $c_\mathrm{o} = q^{2g+3/2}$ and $\eta^2 = 1$, it equals

$$\eta^g s^{-g}\frac{q}{2}(1+\eta q^2s^3)\left(\frac{(q-1)s}{1-s^2}+\sqrt{q}\,\eta\right). \tag{4.49}$$

Substituting (4.49) into (4.48) and cancelling the factor $1+\eta q^2 s^3$ against $1-q^4s^6=(1-\eta q^2s^3)(1+\eta q^2s^3)$, we obtain

$$\mathcal{S}_0(g)=\sum_{k=0}^{2}\sum_{\varrho\in\{\pm1\}}\sum_{\eta\in\{\pm1\}}\operatorname{Res}_{s=\varrho s_k}T_\eta(s), \tag{4.50}$$

with T_η as in (4.45). Fix k and ϱ. Among the denominator factors of T_η only $1-\eta q^2s^3$ can vanish at $s=\varrho s_k$, and it does so exactly when $\eta=\varrho$; hence no denominator factor of $T_{-\varrho}$ is zero there, and the Euler product is holomorphic there by Lemma 4.3, so $T_{-\varrho}$ contributes no residue, whereas T_ϱ carries the double-pole factor $(1-\varrho q^2s^3)^{-2}$. Thus (4.50) reduces to

$$\mathcal{S}_0(g)=\sum_{k=0}^{2}\left\{\operatorname{Res}_{s=s_k}T_{+1}(s)+\operatorname{Res}_{s=-s_k}T_{-1}(s)\right\}.$$

Finally, the substitution $s\mapsto -s$ reverses the sign of a residue, and $T_{-1}(-s)=-T_{+1}(s)$ by (4.46), so $\operatorname{Res}_{s=-s_k}T_{-1}=\operatorname{Res}_{s=s_k}T_{+1}$; substituting this into the display above gives (4.47). □

We now evaluate the representative residues in (4.47). Fix $k\in\{0,1,2\}$. Since $s_k^3=q^{-2}$, one has $q^2z_k=q^2s_k^2=s_k^{-1}$. Consequently,

$$s_k^{-g}=(q^2z_k)^g,\qquad 1-q^{3/2}s_k^3=1-q^{-1/2}. \tag{4.51}$$

Write $\mathscr{D}(z):=\prod_{P\in\mathcal{P}}\mathcal{D}_P(z,qz)$. The residue calculation requires $\mathscr{D}'$ at z_0, z_1 and z_2, so we record that $\mathscr{D}$ is holomorphic there. Each z_k has $|z_k|=q^{-4/3}$ and so lies in the annulus $q^{-2}<|z|<q^{-1}$, on which $w'=qz$ was seen in §4.2.3 to satisfy all five conditions of (4.11); by Lemma 4.3 the product converges absolutely and uniformly on compact subsets of that annulus, and $\mathscr{D}$ is therefore holomorphic near each of z_0, z_1 and z_2.

For $z^3=q^{-4}$, that is, for $z\in\{z_0,z_1,z_2\}$, define

$$\mathfrak{a}(z):=-\frac{(1-q^3z^2)\big[q(q-1)z^2+q^{-1/2}(1-z)\big]}{18z(1-q^2z)(1-z)}\,\mathscr{D}(z) \tag{4.52}$$

and

$$\begin{aligned}\mathfrak{b}(z):=&\left[2+\frac{4q^3z^2}{1-q^3z^2}-\frac{2q^2z}{1-q^2z}-\frac{3}{\sqrt{q}-1}\right]\mathfrak{a}(z)+\frac{q(q-1)z(1+z)(1-q^3z^2)}{18(1-q^2z)(1-z)^2}\,\mathscr{D}(z)\\&+\frac{(1-q^3z^2)\big[q(q-1)z^2+q^{-1/2}(1-z)\big]}{9(1-q^2z)(1-z)}\,\mathscr{D}'(z).\end{aligned} \tag{4.53}$$

None of the denominator factors in (4.52) and (4.53) vanishes at z_0, z_1 or z_2, so both expressions are well defined there.

By (4.51) the factor $1-q^{3/2}s^3$ of (4.45) takes the same nonzero value $1-q^{-1/2}$ at s_0, s_1, s_2; the next lemma clears it by evaluating $(1-q^{-1/2})\operatorname{Res}_{s=s_k}T_{+1}(s)$.

Lemma 4.10 (Residue at a lifted secondary point)**.** *For every $k\in\{0,1,2\}$, one has*

$$(1-q^{-1/2})\operatorname{Res}_{s=s_k}T_{+1}(s)=(q^2z_k)^g\left(\mathfrak{a}(z_k)g+\mathfrak{b}(z_k)\right). \tag{4.54}$$

Proof. Put

$$R(s):=\frac{q(1-q^3s^4)}{2(1-q^2s^2)(1-q^{3/2}s^3)}\left(\frac{(q-1)s}{1-s^2}+\sqrt{q}\right),\qquad \Delta(s):=1-q^2s^3,$$

so that $T_{+1}(s)=G(s)\Delta(s)^{-2}$ with $G(s):=s^{-g}R(s)\mathscr{D}(s^2)$. Here G is holomorphic near s_k, since $\mathscr{D}$ is holomorphic near z_k as shown above and no denominator factor of R vanishes at s_k. Since $q^2s_k^3=1$ we have $\Delta'(s_k)=-3/s_k$ and $\Delta''(s_k)=-6/s_k^2$; as Δ has a simple zero at s_k, expanding G and Δ in powers of $s-s_k$ gives

$$\operatorname{Res}_{s=s_k}T_{+1}(s)=\frac{G'(s_k)}{\Delta'(s_k)^2}-\frac{G(s_k)\Delta''(s_k)}{\Delta'(s_k)^3}=\frac{s_k^2}{9}G'(s_k)-\frac{2s_k}{9}G(s_k).$$

By the product rule, $sG'(s)=s^{-g}\big[(-gR(s)+sR'(s))\mathscr{D}(s^2)+2s^2R(s)\mathscr{D}'(s^2)\big]$, so that, using $s_k^{-g}=(q^2z_k)^g$ and $s_k^2=z_k$,

$$(1-q^{-1/2})\operatorname{Res}_{s=s_k}T_{+1}(s)=(q^2z_k)^g\frac{(1-q^{-1/2})s_k}{9}\Big[\big(-(g+2)R(s_k)+s_kR'(s_k)\big)\mathscr{D}(z_k)+2z_kR(s_k)\mathscr{D}'(z_k)\Big]. \tag{4.55}$$

Using $s_k^2=z_k$, $s_k^3=q^{-2}$ and $z_k^3=q^{-4}$ in the definition of R gives

$$\frac{(1-q^{-1/2})s_k}{9}R(s_k)\mathscr{D}(z_k)=-\mathfrak{a}(z_k). \tag{4.56}$$

Splitting $-(g+2)R(s_k)=-gR(s_k)-2R(s_k)$, the bracketed expression in (4.55) is

$$-gR(s_k)\mathscr{D}(z_k)+\big(s_kR'(s_k)-2R(s_k)\big)\mathscr{D}(z_k)+2z_kR(s_k)\mathscr{D}'(z_k),$$

and we multiply each of the three summands by $(1-q^{-1/2})s_k/9$. The first gives $g\mathfrak{a}(z_k)$ by (4.56). The third is again immediate from (4.56):

$$\frac{(1-q^{-1/2})s_k}{9}\,2z_kR(s_k)\mathscr{D}'(z_k) = -2z_k\mathfrak{a}(z_k)\frac{\mathscr{D}'(z_k)}{\mathscr{D}(z_k)},$$

which is the third summand of (4.53). For the second, differentiating R and using the same relations, a direct computation shows that $\frac{(1-q^{-1/2})s_k}{9}\big(s_kR'(s_k)-2R(s_k)\big)\mathscr{D}(z_k)$ equals the bracketed factor in the first summand of (4.53) multiplied by $\mathfrak{a}(z_k)$, plus the second summand of (4.53). The second and third summands therefore add up to $\mathfrak{b}(z_k)$, and (4.55) becomes (4.54). □

For $m\in\mathbb{Z}$, put

$$a_m := \frac{2}{q^2}\sum_{k=0}^{2} e^{2\pi ikm/3}\mathfrak{a}(z_k), \qquad b_m := \frac{2}{q^2}\sum_{k=0}^{2} e^{2\pi ikm/3}\mathfrak{b}(z_k), \tag{4.57}$$

so that $a_{m+3}=a_m$ and $b_{m+3}=b_m$ by definition, since $e^{2\pi ik(m+3)/3}=e^{2\pi ikm/3}$.

Corollary 4.11. *The completed secondary contribution $\mathcal{S}_0(g)$ satisfies*

$$\mathcal{S}_0(g) = \frac{q^{2g/3+2}}{1-q^{-1/2}}\,(a_gg+b_g)\,, \tag{4.58}$$

and the coefficients a_m, b_m are real for every $m\in\mathbb{Z}$.

Proof. Combining (4.47) with Lemma 4.10 and using $q^2z_k = q^{2/3}e^{2\pi ik/3}$, we obtain

$$\mathcal{S}_0(g) = \frac{2}{1-q^{-1/2}}\sum_{k=0}^{2}(q^2z_k)^g\big(\mathfrak{a}(z_k)g+\mathfrak{b}(z_k)\big) = \frac{2q^{2g/3}}{1-q^{-1/2}}\sum_{k=0}^{2}e^{2\pi ikg/3}\big(\mathfrak{a}(z_k)g+\mathfrak{b}(z_k)\big).$$

By (4.57) the last sum equals $\frac{1}{2}q^2(a_gg+b_g)$, which is (4.58).

The coefficients are real because the local factors $\mathcal{D}_P(z,qz)$ and the rational functions in (4.52) and (4.53) have real coefficients, so that $\mathfrak{a}(\overline{z})=\overline{\mathfrak{a}(z)}$ and $\mathfrak{b}(\overline{z})=\overline{\mathfrak{b}(z)}$: since z_0 is real and $z_2=\overline{z_1}$, the terms with $k=1$ and $k=2$ in (4.57) are complex conjugates and the term with $k=0$ is real. □

Lemma 4.12. *Let $z^3=q^{-4}$. Then $\mathfrak{a}(z)\neq 0$.*

Proof. We first show that $\mathscr{D}(z)\neq 0$. Substituting $w'=qz$ into (4.9) and using (4.7),

$$\mathcal{D}_P(z,qz) = \frac{|P|-1}{|P|}\,\frac{-|P|^4z^{4d(P)}+|P|^3z^{2d(P)}+|P|^2z^{d(P)}-1}{\big(|P|^2z^{d(P)}-1\big)\big(1+|P|z^{d(P)}\big)}. \tag{4.59}$$

Put $X:=z^{d(P)}$, so that $X^3=|P|^{-4}$ since $z^3=q^{-4}$. Neither denominator factor of (4.59) vanishes, and the numerator becomes $|P|^3X^2+(|P|^2-1)X-1$, a quadratic with real coefficients and discriminant $(|P|^2-1)^2+4|P|^3>0$, hence with real roots only. If X is nonreal it is not a root; if X is real then $X^3=|P|^{-4}>0$ forces $X=|P|^{-4/3}$, where the quadratic equals $\big(|P|^{2/3}-1\big)+\big(|P|^{1/3}-|P|^{-4/3}\big)>0$. Thus $\mathcal{D}_P(z,qz)\neq 0$ for every P. As $|z|=q^{-4/3}$ and $|qz|=q^{-1/3}$, the pair (z,qz) satisfies (4.11), so Lemma 4.3 gives $\sum_P\big|\mathcal{D}_P(z,qz)-1\big|<\infty$, and an absolutely convergent product of nonzero factors is nonzero.

It remains to treat the rational prefactor in (4.52), whose numerator factor $1-q^3z^2$ is nonzero as already noted; only $q(q-1)z^2+q^{-1/2}(1-z)$ has to be considered. There the second summand dominates, since $\big|q(q-1)z^2\big| = q^{-2/3}-q^{-5/3}$ while $\big|q^{-1/2}(1-z)\big|\geq q^{-1/2}-q^{-11/6}$, and the difference of these is $\big(q^{-2/3}+q^{-11/6}\big)(q^{1/6}-1)>0$. Hence $q(q-1)z^2+q^{-1/2}(1-z)\neq 0$. Since $1-q^3z^2\neq 0$ and $\mathscr{D}(z)\neq 0$, all three factors in the numerator of (4.52) are nonzero, and therefore $\mathfrak{a}(z)\neq 0$. □

The poles of T_{+1} at s_0, s_1 and s_2 have order exactly two. Indeed, in the notation of the proof of Lemma 4.10, $T_{+1}(s)=G(s)(1-q^2s^3)^{-2}$ with G holomorphic at s_k, and $G(s_k)$ is a nonzero multiple of $\mathfrak{a}(z_k)$ by (4.56), while $\mathfrak{a}(z_k)\neq 0$ by Lemma 4.12; by (4.46) the same holds for T_{-1} at $s=-s_k$. All six surviving secondary points are therefore genuine double poles.

Corollary 4.13. *The sequence $m\mapsto(a_m,b_m)$ has minimal period exactly three.*

Proof. Put $\omega=e^{2\pi i/3}$, so that $\sum_{m=0}^{2}\omega^{(k-1)m}$ is 3 for $k=1$ and 0 for $k\in\{0,2\}$. By (4.57),

$$\sum_{m=0}^{2}a_m\omega^{-m} = \frac{6}{q^2}\,\mathfrak{a}(z_1)\neq 0$$

by Lemma 4.12. Hence a_0, a_1, a_2 are not all equal, so $m\mapsto(a_m,b_m)$ is nonconstant; being periodic with period three, its minimal period cannot be one and is therefore exactly three. □

We now combine the evaluations of the principal and secondary square-dual contributions.

Proof of Proposition 4.1. Summing (4.26) over $N \in \{g, g-1\}$ and using (4.1),

$$\mathcal{S}^*(V=\square) = \sum_{N\in\{g,g-1\}} \big(A^*_{N,\mathrm{e}} + A^*_{N,\mathrm{o}}\big) + \sum_{N\in\{g,g-1\}} \big(B^*_{N,\mathrm{e}} + B^*_{N,\mathrm{o}}\big) + O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big).$$

The first sum equals $-\mathcal{R}_0(g;\rho)$ by Proposition 4.6, and the second equals $\mathcal{S}_0(g) + O_\varepsilon(q^{g(1+\varepsilon)/2})$ by Proposition 4.8. This is (4.2), while Corollary 4.11 gives the stated explicit formula for $\mathcal{S}_0(g)$ and the reality of its coefficients. □

5. Error from nonsquare dual variables

The purpose of this section is to bound the nonsquare-dual contribution $\mathcal{S}^*(V \neq \square)$ of (2.36), the last of the three terms in that decomposition. Unlike the other two, it contributes no main or secondary term: it is $O_\varepsilon(q^{g(1+\varepsilon)/2})$, the size of the error already incurred in (2.36).

Proposition 5.1. *For every $\varepsilon > 0$, one has*

$$\mathcal{S}^*(V \neq \square) = O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big). \tag{5.1}$$

Fix $N \in \{g, g-1\}$. Retaining in (2.30) and (2.31) only the nonsquare dual variables gives

$$\begin{aligned} s_{\mathrm{e}}(h;C)_{\neq\square} = (q-1) \sum_{\substack{V\in\mathcal{A}^+_{\leq\lambda-4}\\ V\neq\square}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} - \sum_{V\in\mathcal{A}^+_{\lambda-3}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} \\ - \frac{q-1}{q} \sum_{\substack{V\in\mathcal{A}^+_{\leq\lambda-2}\\ V\neq\square}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} + \frac{1}{q} \sum_{V\in\mathcal{A}^+_{\lambda-1}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} \end{aligned} \tag{5.2}$$

and

$$s_{\mathrm{o}}(h;C)_{\neq\square} = \sum_{\substack{V\in\mathcal{A}^+_{\lambda-3}\\ V\neq\square}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} - \frac{1}{q} \sum_{\substack{V\in\mathcal{A}^+_{\lambda-1}\\ V\neq\square}} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}}, \tag{5.3}$$

where $\lambda = \lambda(h,C) = d(h) - 2g + 2d(C)$ as in (2.29). The two exact-degree sums of (5.2) carry no restriction because, $d(h)$ being even, their degrees $\lambda-3$ and $\lambda-1$ are odd, whereas a square polynomial has even degree. Throughout this section we call the four dual sums of (5.2) and the two of (5.3) the *six dual sums*.

5.1. A fixed nonsquare dual variable.

For a fixed nonsquare $V \in \mathcal{A}^+$, set $\sigma_V := (-1)^{\frac{q-1}{2}d(V)}$. For $(V,h)=1$,

$$\left(\frac{V}{h}\right) = (-1)^{\frac{q-1}{2}d(V)d(h)} \left(\frac{h}{V}\right) = \sigma_V^{d(h)}\psi_V(h); \tag{5.4}$$

when $(V,h) \neq 1$ both symbols vanish, so (5.4) holds for every $h \in \mathcal{A}^+$. The sign σ_V is identically 1 when $q \equiv 1 \bmod 4$, as assumed in [6], but not for a general odd prime power q.

In (2.32) and (2.34) the variables h and C are entangled by the condition $C \mid h^\infty$. The generating function

$$\sum_{\substack{C\in\mathcal{A}^+\\ C\mid h^\infty}} u^{d(C)} = \prod_{P\mid h} \big(1-u^{d(P)}\big)^{-1}, \qquad (|u|<1), \tag{5.5}$$

is a function of h alone. Accordingly, for $n \geq 0$ and $|u| < 1$, define

$$\Sigma_n(V;u) := \sum_{h\in\mathcal{A}^+_n} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} \prod_{P\mid h} \Big(1-u^{d(P)}\Big)^{-1}. \tag{5.6}$$

Since ψ_V may be imprimitive, we record a Lindelöf-type bound valid for an arbitrary nonprincipal character. In [6, Section 7] this is met instead by writing $V = AD^2$ with D square-free and reducing to a square-free modulus, a reduction that Lemma 5.2 makes unnecessary.

Lemma 5.2 (Function-field Lindelöf bound)**.** *Let χ be a nonprincipal Dirichlet character over $\mathcal{A}$ of modulus $F \in \mathcal{A}^+$. For every $\theta > 0$, uniformly over all such pairs (F,χ) and all ξ satisfying $|\xi| \leq q^{-1/2}$, one has*

$$\mathcal{L}(\xi,\chi) \ll_\theta |F|^\theta. \tag{5.7}$$

Proof. Let χ^* be the primitive character inducing χ, of conductor $F^* \mid F$. The explicit bound of [4, Theorem 20], stated there for the ℓ-th power residue symbol modulo a place, uses the character only through the Riemann hypothesis for the associated primitive L-function and the degree of the conductor, and therefore gives $\mathcal{L}(\xi,\chi^*) \ll_\alpha |F^*|^\alpha$ for every $\alpha>0$ and every ξ with $|\xi| \le q^{-1/2}$; if χ^* is even, the trivial factor $1-\xi$ is bounded by $1+q^{-1/2}$ there and does not affect this. Finally,

$$\mathcal{L}(\xi,\chi) = \mathcal{L}(\xi,\chi^*) \prod_{\substack{P\mid F\\ P\nmid F^*}} \bigl(1-\chi^*(P)\xi^{d(P)}\bigr),$$

and each factor of the product has modulus at most 2 there, so the product is at most $2^{\omega(F)}$, hence at most the number of monic divisors of F, which is $\ll_\alpha |F|^\alpha$. Taking $\alpha=\theta/2$ and using $|F^*|\le|F|$ gives (5.7). □

The following bound on $\Sigma_n(V;u)$ is [6, (7.5)] in the twisted form required here. Its proof follows [6, Section 7], and indicates where the twist and the imprimitivity of ψ_V enter.

Lemma 5.3. *Let $\beta,\delta>0$. Uniformly for $|u|=q^{-\beta}$, $n\ge 0$, and nonsquare $V\in\mathcal{A}^+$, one has*

$$\Sigma_n(V;u) \ll_{\beta,\delta} q^{n(1/2+\beta)}|V|^\delta. \tag{5.8}$$

Proof. Let u be any point with $|u|=q^{-\beta}$; every implied constant below depends only on β and δ, which gives the uniformity in u asserted. Put

$$\mathcal{F}(V;w,u) := \sum_{h\in\mathcal{A}^+} \frac{\mathcal{G}(V,\psi_h)}{\sqrt{|h|}} w^{d(h)} \prod_{P\mid h} \bigl(1-u^{d(P)}\bigr)^{-1},$$

whose coefficient of w^n is $\Sigma_n(V;u)$; this is the twisted counterpart of the series $B(V;w,u)$ of [6, (7.2)]. By (2.19) its coefficients are multiplicative in h, and since the factor $(1-u^{d(P)})^{-1}$ occurs exactly when $P\mid h$, its P-Euler factor is

$$1+\frac{1}{1-u^{d(P)}}\sum_{r\ge1}\frac{\mathcal{G}(V,\psi_{P^r})}{|P|^{r/2}}w^{rd(P)}. \tag{5.9}$$

For $P\nmid V$ only $r=1$ survives in (2.18), and (5.4) gives $\mathcal{G}(V,\psi_P)/|P|^{1/2} = \left(\frac{V}{P}\right) = \sigma_V^{d(P)}\psi_V(P)$, so that (5.9) becomes

$$1+\frac{\psi_V(P)(\sigma_V w)^{d(P)}}{1-u^{d(P)}}. \tag{5.10}$$

This is the one place at which the twist enters: [6, (7.2)] has the same factor with w in place of $\sigma_V w$. For $P\mid V$, writing $V=P^kV_1$ with $P\nmid V_1$, the terms with $r\ge k+2$ vanish by (2.18), so (5.9) is a polynomial in w. Since $|1-u^{d(P)}|^{-1}\le(1-q^{-\beta})^{-1}$ and $\sum_P |w|^{d(P)}$ converges for $|w|<q^{-1}$, the series $\mathcal{F}$ converges absolutely there and admits an Euler product.

Cauchy's formula will be applied on $|w|=r_0:=q^{-1/2-\beta}$, chosen because $r_0^{-n}=q^{n(1/2+\beta)}$; as the circle $|w|=r_0$ may lie outside the disc $|w|<q^{-1}$ of absolute convergence, $\mathcal{F}$ must first be continued, which is done by the repeated Euler-factor extraction of [6, (7.3)]. Choose an integer $J\ge1$ with $(J+1)\beta>1/2$, so that $r_0|u|^J = q^{-1/2-(J+1)\beta}<q^{-1}$; such a J depends only on β. As $r_0^2=q^{-1-2\beta}<q^{-1}$ as well, both inequalities persist for some $r_1>r_0$, that is, $r_1|u|^J<q^{-1}$ and $r_1^2<q^{-1}$. For $P\nmid V$, put

$$R_{P,J}(V;w,u) := \left(1+\frac{\psi_V(P)(\sigma_V w)^{d(P)}}{1-u^{d(P)}}\right)\prod_{j=0}^{J-1}\Bigl(1-\psi_V(P)(\sigma_V w u^j)^{d(P)}\Bigr), \tag{5.11}$$

so that (5.10) equals $\Bigl(\prod_{j<J}\bigl(1-\psi_V(P)(\sigma_V wu^j)^{d(P)}\bigr)^{-1}\Bigr)R_{P,J}$. Writing $X:=\psi_V(P)(\sigma_V w)^{d(P)}$ and $t:=u^{d(P)}$, we have $R_{P,J}=\bigl(1+\frac{X}{1-t}\bigr)\prod_{j<J}(1-Xt^j)$; substituting $\frac{1}{1-t}=\sum_{j<J}t^j+\frac{t^J}{1-t}$ shows that the coefficient of X^0 is 1 and that of X^1 is $t^J/(1-t)$, the terms $t^0,\dots,t^{J-1}$ being exactly what the extracted factors cancel. As $|t|\le q^{-\beta}$, the polynomial $R_{P,J}$ has degree $J+1$ in X with coefficients bounded in terms of β, and $|X|\le|w|^{d(P)}$; hence

$$R_{P,J}(V;w,u) = 1+O_\beta\bigl(|wu^J|^{d(P)}+|w|^{2d(P)}\bigr)\qquad(|w|\le r_1). \tag{5.12}$$

Multiplying (5.11) over $P\nmid V$ and adjoining the finitely many factors (5.9) at $P\mid V$, and using $\psi_V(P)=0$ for $P\mid V$ to complete each extracted product to a full Euler product, we obtain

$$\mathcal{F}(V;w,u) = \Bigl(\prod_{j=0}^{J-1}\mathcal{L}(\sigma_V wu^j,\psi_V)\Bigr)\mathcal{C}_J(V;w,u), \tag{5.13}$$

where $\mathcal{C}_J$ denotes the product of the $R_{P,J}$ over $P\nmid V$ and of the local factors (5.9) over $P\mid V$. By (5.12) and the inequalities $r_1|u|^J<q^{-1}$ and $r_1^2<q^{-1}$, the series $\sum_P\bigl((r_1|u|^J)^{d(P)}+r_1^{2d(P)}\bigr)$ converges, so $\prod_{P\nmid V}R_{P,J}$

converges absolutely and uniformly for $|w| \le r_1$ and is $\ll_\beta 1$ there, uniformly in V. Since each $\mathcal{L}(\sigma_V w u^j, \psi_V)$ is a polynomial in w and the remaining product is finite, the right-hand side of (5.13) is holomorphic for $|w| < r_1$ and, by the identity theorem, continues $\mathcal{F}(V; w, u)$ there; in particular (5.13) holds on $|w| = r_0$.

On $|w| = r_0$ one has $|w|^{d(P)} = |P|^{-1/2-\beta}$, so by (2.18) the r-th term of (5.9) has modulus at most $|P|^{-\beta r}$ when $r \le k$ is even, at most $|P|^{-1/2-\beta r}$ when $r = k+1$, and vanishes otherwise. Summing over r and using $|1-u^{d(P)}|^{-1} \le (1-q^{-\beta})^{-1}$ bounds each factor at $P \mid V$ by a constant C_β, uniformly in P and k; their product is therefore at most $C_\beta^{\omega(V)}$; since $2^{\omega(V)}$ is at most the number of monic divisors of V, which is $\ll_\alpha |V|^\alpha$ for every $\alpha > 0$, this is $\ll_{\beta,\delta} |V|^{\delta/2}$. Consequently $\mathcal{C}_J(V; w, u) \ll_{\beta,\delta} |V|^{\delta/2}$ on $|w| = r_0$.

Finally, ψ_V is nonprincipal by (2.1), V not being a square, and on $|w| = r_0$ one has $|\sigma_V w u^j| = q^{-1/2-(j+1)\beta} < q^{-1/2}$ for $0 \le j < J$, so Lemma 5.2 with $F = V$ and $\theta = \delta/(2J)$ gives $\prod_{j<J} \mathcal{L}(\sigma_V w u^j, \psi_V) \ll_{\beta,\delta} |V|^{\delta/2}$. Hence $\mathcal{F}(V; w, u) \ll_{\beta,\delta} |V|^\delta$ on $|w| = r_0$, and estimating $\Sigma_n(V; u) = \frac{1}{2\pi i} \oint_{|w|=r_0} \mathcal{F}(V; w, u) w^{-n-1}\, dw$ trivially, with $r_0^{-n} = q^{n(1/2+\beta)}$, gives (5.8). □

5.2. **Completion of the nonsquare estimate.** We now combine the fixed-V coefficient bound of Lemma 5.3 with the six dual sums to prove Proposition 5.1.

Proof of Proposition 5.1. All $C \in \mathcal{A}_j^+$ have $|C| = q^j$, and by (5.5) the number of those dividing h^∞ is the coefficient of u^j in $\prod_{P|h}(1-u^{d(P)})^{-1}$. Cauchy's formula therefore gives, for $j \ge 0$ and any $\beta > 0$, the identity of [6, Section 7],

$$\sum_{\substack{C \in \mathcal{A}_j^+ \\ C | h^\infty}} \frac{1}{|C|^2} = q^{-2j} \#\{C \in \mathcal{A}_j^+ : C \mid h^\infty\} = \frac{q^{-2j}}{2\pi i} \oint_{|u|=q^{-\beta}} \frac{1}{u^{j+1}} \prod_{P|h} \left(1 - u^{d(P)}\right)^{-1} du. \tag{5.14}$$

Fix $n \ge 0$ and a nonsquare $V \in \mathcal{A}_\ell^+$. Substituting (5.14) into the h-sum of degree n and interchanging the finite sums with the integral, (5.6) and Lemma 5.3 give

$$\sum_{h \in \mathcal{A}_n^+} \frac{\mathcal{G}(V, \psi_h)}{\sqrt{|h|}} \sum_{\substack{C \in \mathcal{A}_j^+ \\ C | h^\infty}} \frac{1}{|C|^2} = \frac{q^{-2j}}{2\pi i} \oint_{|u|=q^{-\beta}} \frac{\Sigma_n(V; u)}{u^{j+1}}\, du \ll_{\beta,\delta} q^{-2j+\beta j} q^{n(1/2+\beta)} |V|^\delta, \tag{5.15}$$

the second step because the contour length $2\pi q^{-\beta}$ and the factor $|u|^{-(j+1)} = q^{\beta(j+1)}$ together contribute $q^{\beta j}$.

Since $\#\mathcal{A}_\ell^+ = q^\ell$ and $|V| = q^\ell$ for $V \in \mathcal{A}_\ell^+$, summing the moduli of the left-hand sides of (5.15) over the nonsquare $V \in \mathcal{A}_\ell^+$ yields

$$\sum_{\substack{V \in \mathcal{A}_\ell^+ \\ V \ne \square}} \left| \sum_{h \in \mathcal{A}_n^+} \frac{\mathcal{G}(V, \psi_h)}{\sqrt{|h|}} \sum_{\substack{C \in \mathcal{A}_j^+ \\ C | h^\infty}} \frac{1}{|C|^2} \right| \ll_{\beta,\delta} q^{-2j+\beta j} q^{n(1/2+\beta)} q^{\ell(1+\delta)}. \tag{5.16}$$

For $L \ge 0$, summing (5.16) over $0 \le \ell \le L$ and using $\sum_{\ell=0}^{L} q^{\ell(1+\delta)} \ll_\delta q^{L(1+\delta)}$, we obtain

$$\sum_{\substack{V \in \mathcal{A}^+ \\ d(V) \le L \\ V \ne \square}} \left| \sum_{h \in \mathcal{A}_n^+} \frac{\mathcal{G}(V, \psi_h)}{\sqrt{|h|}} \sum_{\substack{C \in \mathcal{A}_j^+ \\ C | h^\infty}} \frac{1}{|C|^2} \right| \ll_{\beta,\delta} q^{-2j+\beta j} q^{n(1/2+\beta)} q^{L(1+\delta)}. \tag{5.17}$$

We now apply (5.16) to the dual sums of exact degree and (5.17) to those of bounded degree. Recall that $N \in \{g, g-1\}$; fix $n = d(h)$ and $j = d(C)$, so that $\lambda = n - 2g + 2j$ by (2.29), and for each of the six dual sums let μ denote its exact dual degree or, in the two cumulative cases, its upper bound; the six values are listed in the third column of the table below. Applying (5.16) or (5.17) with $\ell = \mu$ or $L = \mu$ bounds the term, at fixed n and j, by

$$\Big(\sum_{\substack{m \in \mathcal{A}^+ \\ d(m) \le N-n}} |m|^{-3/2} \Big) q^{\beta(n+j)+\delta\mu} \times \text{(last column below)},$$

where the factor $q^{\beta(n+j)+\delta\mu}$ is common to all six terms. The last column collects the remaining powers of q, that is, the prefactor q^{2g+2} or $q^{2g+5/2}$, the scalar factor, and the powers q^{-n}, q^{-2j}, $q^{n/2}$ and q^μ supplied by (5.16) and (5.17):

term	scalar factor	dual degree or upper bound	last column
even, first	$q-1$	$d(V)\le\lambda-4$	$q^{n/2-1}$
even, second	-1	$d(V)=\lambda-3$	$q^{n/2-1}$
even, third	$-(q-1)/q$	$d(V)\le\lambda-2$	$q^{n/2}$
even, fourth	$1/q$	$d(V)=\lambda-1$	$q^{n/2}$
odd, first	1	$d(V)=\lambda-3$	$q^{n/2-1/2}$
odd, second	$-1/q$	$d(V)=\lambda-1$	$q^{n/2+1/2}$

A dual sum with $\mu<0$ is empty, so we may assume $\mu\ge0$; the largest of the six values of μ is $\lambda-1$, which is at most $n-1$ since $j\le g$, and $d(h)+d(m)\le N$ forces $n\le N\le g$. Hence $\beta(n+j)\le2\beta g$ and $\delta\mu\le\delta g$, so the common factor is at most $q^{(2\beta+\delta)g}$. Moreover $\sum_{d(m)\le N-n}|m|^{-3/2}\le\big(1-q^{-1/2}\big)^{-1}$.

Every entry of the last column is $O_q(q^{n/2})$. Taking absolute values in (2.32) and (2.34), inserting the six bounds together with the two estimates just proved, and summing over $0\le j\le g$ and $0\le n\le N$ – which only enlarges the bound, each sector restricting n to one parity – we obtain, for $\bullet\in\{\mathrm{e},\mathrm{o}\}$,

$$\mathcal{S}^{*}_{N,\bullet}(V\neq\square)\ll_{\beta,\delta}q^{(2\beta+\delta)g}\sum_{j=0}^{g}\sum_{n=0}^{N}q^{n/2}\ll_{\beta,\delta}(g+1)\,q^{N/2+(2\beta+\delta)g}.$$

Given $\varepsilon>0$, choose $\beta,\delta>0$ with $2\beta+\delta<\varepsilon/4$. Since $g+1\ll_\varepsilon q^{\varepsilon g/4}$ and $N\le g$, the right-hand side is $\ll_\varepsilon q^{N/2+\varepsilon g/2}\ll_\varepsilon q^{g(1+\varepsilon)/2}$. This bounds $\mathcal{S}^{*}_{N,\mathrm{e}}(V\neq\square)$ and $\mathcal{S}^{*}_{N,\mathrm{o}}(V\neq\square)$ for each $N\in\{g,g-1\}$, so (2.39) gives (5.1). □

6. Proof of Theorem 1.1

We assemble the three contributions evaluated in §3, §4 and §5: the two retained contours cancel, which evaluates $M^{*}(g)$, and (1.3) returns the result to $M(g)$.

Proof of Theorem 1.1. Let $\varepsilon>0$ and fix ρ with $1<\rho<q$. Propositions 3.1 and 4.1 hold for every such ρ; applied with the same ρ they give $\mathcal{M}^{*}=\mathcal{Q}_g+\mathcal{R}_0(g;\rho)+O_\varepsilon(q^{\varepsilon g})$ and $\mathcal{S}^{*}(V=\square)=-\mathcal{R}_0(g;\rho)+\mathcal{S}_0(g)+O_\varepsilon(q^{g(1+\varepsilon)/2})$, so that the two occurrences of $\mathcal{R}_0(g;\rho)$ are the same quantity and cancel on adding. With Proposition 5.1, the decomposition (2.36) therefore gives

$$M^{*}(g)=\mathcal{Q}_g+\mathcal{S}_0(g)+O_\varepsilon\big(q^{\varepsilon g}\big)+O_\varepsilon\big(q^{g(1+\varepsilon)/2}\big).$$

It suffices to prove (1.4) for $0<\varepsilon\le1$, since for larger ε it follows from the case $\varepsilon=1$; for such ε one has $\varepsilon g\le g(1+\varepsilon)/2$, whence

$$M^{*}(g)=\mathcal{Q}_g+\mathcal{S}_0(g)+O_\varepsilon\Big(q^{g(1+\varepsilon)/2}\Big). \tag{6.1}$$

By (1.3), $M(g)=(1-q^{-1/2})M^{*}(g)$, and $0<1-q^{-1/2}<1$ leaves the error term unchanged. In (4.58) the factor $1-q^{-1/2}$ cancels the denominator, so that, with $\mathcal{S}(g)$ as in (1.6),

$$(1-q^{-1/2})\mathcal{S}_0(g)=q^{2g/3+2}\big(a_g g+b_g\big)=\mathcal{S}(g);$$

and by (3.7) $(1-q^{-1/2})\mathcal{Q}_g$ is the principal term $\mathcal{M}_{\mathrm{prin}}(g)$ of (1.5). Multiplying (6.1) by $1-q^{-1/2}$ therefore gives (1.4).

Finally, the six coefficients a_0,a_1,a_2,b_0,b_1,b_2 are real by Corollary 4.11 and do not depend on g, and $m\mapsto(a_m,b_m)$ has minimal period exactly three by Corollary 4.13. □

Remark 6.1 (Comparison with the earlier even-degree formulas). In the argument above the secondary contribution is the sum of the residues at the three zeros of $1-q^4z^3$, namely $z_k=q^{-4/3}e^{2\pi ik/3}$ with $k\in\{0,1,2\}$. They enter the final answer only through the factor $(q^2z_k)^g=q^{2g/3}e^{2\pi ikg/3}$ of Lemma 4.10, and by Lemma 4.12 none of the three residues vanishes identically in g. This is the origin of the period-three behaviour.

At the corresponding step the earlier treatments retain only the residue at the real point $z=q^{-4/3}$: the z-contour is enlarged from $|z|=q^{-3/2}$ to $|z|=q^{-1-\varepsilon}$ across an integrand whose denominator contains $1-q^4z^3$, and the residue is taken there; see [10, Sections 3.2.1 and 3.2.2], [3, Lemma 5.7] and, for the odd-degree family, [6, Section 6]. All three zeros of $1-q^4z^3$ have modulus $q^{-4/3}$ and so lie strictly inside the annulus crossed by that shift; keeping only $z=q^{-4/3}$ suppresses the phases $e^{2\pi ikg/3}$ and yields a lower-order term with no dependence on g mod 3, which is the shape recorded in [10, Theorem 1.1] and [3, Theorem 1.4]. Remark 6.2 shows that the period-three shape is the one realized by $M(g)$.

The simple-pole branch is a second point of difference. Here it is removed by Lemma 4.9 once the two truncation levels $N=g$ and $N=g-1$ are combined, in agreement with [10, (4.5)]; [3, (5.38)] retains from it two further terms, of sizes $q^{g/6+\lfloor g/2\rfloor}$ and $q^{g/6+\lfloor(g-1)/2\rfloor}$, both of order $q^{2g/3}$.

Remark 6.2 (Numerical verification). The moment $M(g)$ can be computed exactly for small q and g: by Lemma 2.2 and (2.24) it is determined by the integers

$$S_n := \sum_{h\in\mathcal{A}_n^+} \sum_{D\in\mathcal{H}_{2g+2}} \chi_D(h), \qquad 0 \le n \le g,$$

and each S_n is obtained from the generating series of Lemma 2.4. With $\mathcal{M}_{\mathrm{prin}}$ as in (1.5), the table below lists $M(g) - \mathcal{M}_{\mathrm{prin}}(g)$, the predicted secondary term $\mathcal{S}(g)$ of (1.6), their ratio, and the normalized remainder $\big(M(g) - \mathcal{M}_{\mathrm{prin}}(g) - \mathcal{S}(g)\big)q^{-g/2}$. The Euler products $\mathscr{P}(1)$, $(\mathscr{P}'/\mathscr{P})(1)$ and $\prod_P \mathcal{D}_P(z_k, qz_k)$ were evaluated from the exact counts $\#\mathcal{P}_d$.

q	g	$M - \mathcal{M}_{\mathrm{prin}}$	$\mathcal{S}(g)$	ratio	$(M - \mathcal{M}_{\mathrm{prin}} - \mathcal{S})q^{-g/2}$
3	6	21.98	19.28	1.140	0.100
3	7	−179.66	−169.78	1.058	−0.211
3	8	191.51	172.01	1.113	0.241
5	4	−76.71	−79.34	0.967	0.105
5	5	155.17	151.43	1.025	0.067
5	6	50.50	53.04	0.952	−0.020
7	4	−291.39	−293.13	0.994	0.036
7	5	664.29	686.72	0.967	−0.173
11	4	−1470.41	−1461.98	1.006	−0.070
13	4	−2585.68	−2568.95	1.007	−0.099

The normalized remainders stay bounded, in agreement with the error term in (1.4). The sign pattern is the point at issue: in every row of the table the quantity $M - \mathcal{M}_{\mathrm{prin}}$ is positive at $g \equiv 0, 2 \bmod 3$ and large negative at $g \equiv 1 \bmod 3$, and for $q \in \{7, 11, 13\}$ the predicted $\mathcal{S}(g)$ reproduces it to within one percent at $g = 4$. A lower-order term of the form $q^{2g/3}\big(c_1 g + c_0 + \pi(g)\big)$ with c_1, c_0 constant and π of period two would force $q^{-2(g+2)/3}\big(M - \mathcal{M}_{\mathrm{prin}}\big)(g+2) - q^{-2g/3}\big(M - \mathcal{M}_{\mathrm{prin}}\big)(g)$ to be asymptotically constant; for $q = 7$ the values of this difference at $g = 1, 2, 3$ are 0.098, −2.487 and 0.469, which is not consistent with any such shape.

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